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A Generative AI system to improve detection of lesions in MRIs of Liver

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Declaration

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Abstract

The development of robust medical AI systems is often limited by data scarcity, primarily due to privacy regulations and the labor-intensive nature of expert annotation. This study introduces a three-dimensional denoising diffusion probabilistic model (3D-DDPM) for generating synthetic hepatic MRI volumes. The proposed framework utilizes T2-weighted liver MRIs spanning seven lesion categories and employs an unconditional 3D U-Net architecture to iteratively transform Gaussian noise into anatomically coherent volumes through reverse denoising. Model training incorporated mixed-precision computation, gradient accumulation, and bias monitoring to ensure stability and preserve high-resolution synthesis capabilities. Evaluation was conducted using structural similarity (SSIM), peak signal-to-noise ratio (PSNR), and distributional analysis of intensity profiles and lesion features. The synthetic volumes exhibited structurally coherent liver structures and lesion characteristics with contrast dynamics consistent with clinical imaging. This method offers a privacy-preserving solution for medical data augmentation, with potential applications in algorithm development, creating educational resources, and supporting clinical research where access to real patient data is restricted. The findings advance generative modeling in medical imaging and address critical challenges in data availability for healthcare AI systems.

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Chapter 1

Introduction

Liver cyst segmentation is a critical component of radiological and surgical planning workflows, contributing to the diagnosis, monitoring, and treatment of various hepatic pathologies [14]. Hepatic cysts are frequently discovered in abdominal imaging, with most being benign and asymptomatic [14]. However, complex cysts such as biliary cystadenomas, cystadenocarcinomas, hydatid cysts, and cystic metastases can present diagnostic ambiguity and may indicate malignancy, thereby necessitating precise identification and characterization [69].

Automated segmentation methods offer the potential for high-throughput and objective analysis of cystic structures, enabling volumetric quantification, progression tracking, and patient-specific treatment planning [53]. Nevertheless, building robust segmentation systems remains challenging due to the heterogeneity of cyst morphology, overlapping intensity patterns in MR imaging, and the scarcity of annotated datasets [72].

This dissertation investigates the development of a three-dimensional Denoising Diffusion Probabilistic Model (DDPM3D) designed to generate high-quality synthetic T2-weighted MRI volumes of the liver. By augmenting limited training data with structurally coherent and privacy-preserving synthetic examples, the proposed system aims to improve hepatic cyst segmentation performance while advancing the utility of generative models in clinical imaging applications.

1.1 Background

Hepatic cysts are fluid-filled lesions that arise within the liver parenchyma [14] and are frequently identified during routine abdominal imaging. Reported prevalence rates vary, typically ranging from 2.5% to 18%, depending on the imaging modality and demographics of the population studied [14, 10]. Simple cysts are generally asymptomatic and require no intervention [14]. However, complex variants such as biliary cystadenomas, cystadenocarcinomas, hydatid cysts, and cystic metastases can result in pain, infection, hemorrhage, or pose malignant risk [69].

In conditions like polycystic liver disease (PLD) and autosomal dominant polycystic

kidney disease (ADPKD), patients often present with multiple or diffuse cysts, significantly altering hepatic architecture and increasing clinical complexity [31]. In these cases, cyst burden is strongly associated with hepatomegaly, symptom severity, and diminished quality of life, highlighting the need for precise volumetric assessment [48].

1.1.1 Challenges in Existing Automated Segmentation Approaches

Accurate hepatic cyst segmentation remains technically challenging. Cysts exhibit broad variability in size (from millimetric to massive), multiplicity (single versus hundreds), and morphology (round, lobulated, septated) [14]. On T2-weighted MRI, their hyperintense signal can mimic adjacent bile ducts, vessels, or fluid collections, causing frequent misclassification [14]. Traditional segmentation methods such as thresholding, region growing, or active contouring perform poorly in heterogeneous data due to rigidity and sensitivity to noise [38].

The application of deep learning, especially convolutional neural networks (CNNs), has markedly improved segmentation accuracy in many domains. Frid-Adar et al. [19] demonstrated that augmenting training data with GAN-generated synthetic liver lesion images could boost classification sensitivity from approximately 78.6% to 85.7% and specificity from 88.4% to 92.4%. Yet, segmentation requires anatomically consistent, high-resolution data, a need not met by limited GAN-generated samples [72].

1.1.2 Data Scarcity and Privacy Constraints

Training robust AI models requires large annotated datasets, but medical imaging faces critical constraints due to privacy regulations, institutional barriers, and annotation costs. These limitations create bottlenecks that impede progress in automated medical image analysis.

1.1.3 Generative AI and Diffusion Models as a Solution

Generative AI offers a novel solution to address both data scarcity and privacy concerns [65]. GAN-based augmentation has shown promise in boosting CNN performance for classification tasks; however, GANs often lack training stability and can produce inconsistent anatomical structures [19, 65]. Diffusion models, in contrast, generate images by iteratively refining noise toward a target distribution, improving both stability and fidelity [25]. Recent studies indicate that diffusion-generated images can rival or even surpass the realism of real images based on expert assessment [55, 37]. By implementing a 3D DDPM capable of generating anatomically coherent synthetic MRI volumes, researchers can produce large-scale, privacy-preserving datasets that facilitate robust model training without requiring extensive manual annotation [12]. This approach holds significant promise for transforming data-limited medical imaging domains.

1.2 Aims and Objectives

The aim of this project is to design and implement a 3D Denoising Diffusion Probabilistic Model (DDPM3D) for generating anatomically realistic synthetic liver lesion MRI data, specifically targeting hepatic cyst segmentation. By supplying high-fidelity synthetic imaging, the plan is to improve performance of segmentation models trained on limited real data, while alleviating privacy concerns and data scarcity.

Objectives:

1. **Data Collection and Preprocessing:** Gather liver lesion MRI datasets and apply preprocessing via MONAI, including standardization, normalization, resizing, and augmentation techniques.
2. **3D Diffusion Model Development:** Architect and develop a 3D DDPM tailored to generate high-resolution hepatic MRI volumes.
3. **Model Training and Optimization:** train the 3D diffusion model using memory-efficient strategies including gradient accumulation and mixed precision training. The training process will be optimized to generate structurally coherent liver MRI volumes while managing computational constraints.
4. **Quantitative Image Quality Evaluation:** Evaluate synthetic outputs using structural similarity measures such as SSIM and PSNR, along with distributional analysis to validate anatomical fidelity and intensity characteristics.
5. **Synthetic Data Quality Assessment:** Analyze the fidelity and diversity of generated volumes through comprehensive similarity metrics and visual inspection to determine potential applicability for medical imaging applications.

1.3 Overview of the Report

This dissertation is structured as follows:

- **Chapter 1: Introduction** - Defines problem context, motivation, and research direction.
- **Chapter 2: Literature Survey** - Analyzes prior work in segmentation approaches, data scarcity, generative AI, and diffusion-based image synthesis.
- **Chapter 3: Methodology** - Details dataset preparation, DDPM3D architectural choices, implementation framework, and evaluation methodology.
- **Chapter 4: Results and Analysis** - Presents synthetic image quality, segmentation model performance, and comparative analysis.
- **Chapter 5: Conclusions** - Summarizes contributions, acknowledges limitations, and outlines future directions.

1.4 Relationship Between the Project and Degree Programme

This work is closely aligned with learning objectives of the MSc Artificial Intelligence programme at the University of Sheffield. It integrates knowledge of deep learning architectures, scalable model training, and generative modeling. The implementation demonstrates practical competence with modern AI frameworks such as PyTorch and MONAI. Furthermore, the project addresses real-world challenges such as data scarcity, privacy considerations, and medical applicability exemplifying the interdisciplinary and ethically mindful problem-solving emphasized in the programme.

Chapter 2

Literature Survey

2.1 Overview

This literature review comprehensively examines the domain of hepatic cyst analysis in medical imaging, focusing on (1) clinical significance, (2) automated segmentation strategies, (3) dataset constraints, (4) augmentation techniques, and (5) generative modeling approaches especially three-dimensional denoising diffusion probabilistic models (3D-DDPMs). Hepatic cysts are frequent incidental findings, but the variability in their morphology and the limitations of current imaging data impair accurate detection, classification, and longitudinal tracking. MRI, particularly T2-weighted sequences, offers superior contrast and lesion delineation [14]. Yet segmentation remains challenging when cyst appearance overlaps with vascular or biliary structures. Generative models, particularly 3D-DDPMs, provide a mechanistic pathway to augmenting scarce datasets with synthetic realistic imagery to bolster analysis pipelines.

2.2 Clinical Context of Hepatic Cysts

Hepatic cysts are predominantly benign and often asymptomatic, detected incidentally through imaging. Epidemiological studies report varying prevalence: ultrasound reveals around 3–5%, while CT scans reveal up to 18% in adults [10, 14]. Cysts range from simple benign lesions to complex entities such as biliary cystadenoma, cystadenocarcinoma, hydatid cyst, or cystic metastases, each with differential clinical implications [4]. In polycystic liver disease (PLD) and autosomal dominant polycystic kidney disease (ADPKD), cyst burden causes significant hepatomegaly, markedly impacting patients’ daily function and quality of life [48]. Clinical thresholds guiding intervention often rely on volumetric metrics tied to symptom severity [5]. Classification systems like Gigot and Schnelldorfer aid surgical planning by categorizing PLD based on cyst distribution and remaining healthy parenchyma [31].

2.3 Imaging Modalities in Hepatic Cyst Evaluation

Ultrasound is widely used for initial detection due to its accessibility and cost-effectiveness but lacks sensitivity for subtle or deep-seated lesions [10]. CT provides better resolution and reliable identification of cyst-wall characteristics but exposes patients to radiation, limiting longitudinal use [14]. MRI, particularly T2-weighted sequences, is increasingly regarded as the preferred modality for hepatic cyst characterization. [14]. Simple cysts appear as sharply demarcated hyperintense regions, while complex lesions may exhibit septations, mural nodules, or wall irregularities, indicating possible neoplastic or infectious processes [14]. Multiparametric MRI, with diffusion-weighted and contrast-enhanced sequences, enhances diagnostic specificity but introduces heterogeneity in imaging protocols that reduces model generalizability [14].

2.4 Automated Segmentation of Hepatic Cysts

Automated segmentation plays a critical role in hepatic cyst analysis, particularly for volumetric quantification in disorders such as autosomal dominant polycystic kidney disease (ADPKD) [53] and polycystic liver disease (PLD) [31]. Accurate volumetric assessment enables clinicians to monitor disease progression, evaluate symptom severity, and plan personalized therapeutic interventions.

Clinical Context in PLD : In polycystic liver disease (PLD), objective and reproducible volumetric measurements are not only valuable for monitoring disease progression, but also form the basis of clinical decision-making. Treatment decisions such as the use of somatostatin analogues depend on accurate cyst burden quantification, as these drugs have been shown to slow liver volume growth [66]. Surgical interventions like cyst fenestration or hepatic resection are indicated only when cyst volume and distribution meet specific symptomatic or anatomical thresholds [31]. Furthermore, access to the liver transplant waiting list for PLD patients is governed by MELD exception policies that rely on measurable evidence of disease severity including cystic load [51], and PLD patients face notably longer wait times, suggesting that accurate volumetry is crucial in prioritizing care [3]. Consequently, segmentation methods lacking precision may directly impact treatment eligibility and timely referrals.

Traditional methods laid the groundwork for segmentation [54], but the advent of deep learning has notably transformed medical image analysis more broadly delivering substantial improvements in accuracy, robustness, and scalability [42]. Specifically, in hepatic cyst segmentation (e.g., ADPKD), methods like nnU-Net have demonstrated marked gains in precision and generalizability [53].

2.4.1 Importance of Segmentation and Volumetric Measurement

Accurate segmentation and volumetric assessment form the cornerstone of clinical decision-making, particularly in PLD and ADPKD. In these patients, cystic burden significantly influences hepatomegaly, gastrointestinal symptoms, and declines in physical quality of life [48]. Quantitative metrics such as total liver volume (TLV), total cyst volume (TCV), and cyst–parenchyma surface area (CPSA) offer stronger prognostic value than simple linear measurements. Manual volumetry remains highly accurate but is labor-intensive and subject to inter-observer variability. For instance, fully automated methods can quantify TLV and TKV with accuracy and precision comparable to manual tracing, yet operate much faster and more reproducibly [67, 73]. Automated techniques for TCV segmentation in ADPKD MR images perform within the range of inter-reader variability (e.g., differences of 3.9%–8.0%, versus –2.0% between readers) [39]. Moreover, 3D simulation software for liver volumetry outperforms manual methods in both reproducibility and efficiency [50], while CNN-based liver masks substantially reduce inter-observer inconsistency and segmentation time [15]. Despite these technological advances, AI implementations are still constrained by data availability and imaging heterogeneity.

2.4.2 Traditional Image Processing Techniques

Earlier hepatic cyst segmentation efforts relied heavily on classical image processing techniques such as thresholding, region growing, and active contour models [54]. Kim et al. (2016) developed a semi-automated framework based on spatial prior probability maps (SPPM) combined with adaptive thresholding, achieving shape congruence of 84% and a volumetric correlation coefficient (ICC) of 0.91 on MRI-based cyst segmentation [38]. However, these techniques perform poorly in more complex scenarios, especially when:

- **Complex Morphologies:** Multiloculated cysts with irregular septations often break the core assumptions of threshold-based models.
- **Low Contrast:** Simple thresholding fails where cysts overlap with vessels, ducts, or surrounding parenchyma.
- **Noise Sensitivity:** Region-growing methods are inherently sensitive to seed initialization and imaging artifacts, which often compromise segmentation quality in noisy or heterogeneous scans.

While these methods provided a foundation, their inability to generalize across heterogeneous imaging datasets limited their practical clinical application.

2.4.3 Emergence of Deep Learning-Based Segmentation

The introduction of deep learning architectures has significantly transformed cyst segmentation pipelines. Moghadam et al. [53] applied the nnU-Net framework to 127

T2-weighted MRI scans of autosomal dominant polycystic kidney disease (ADPKD) patients, and validated the model on an external cohort of 30 cases. The model achieved a patient-level Dice similarity coefficient (PDice) of $82\% \pm 11\%$ on internal validation and $80\% \pm 12\%$ on external data, with voxel-level true positive rates (VTPR) of $75\% \pm 15\%$ and $91\% \pm 7\%$, respectively [53].

To assess robustness across repeated acquisitions, the authors evaluated test–retest reproducibility in 17 ADPKD patients. They reported coefficients of variation of 94% for liver cyst fraction and 92% for total cyst count, indicating excellent reproducibility in volumetric biomarker estimation [53].

Unlike classical segmentation approaches, nnU-Net dynamically adapts its architecture to dataset-specific characteristics, removing the need for extensive manual tuning and achieving generalizable performance across institutional settings [29].

Similarly, Li et al. [40] proposed the IRIS interactive segmentation platform for liver cysts in ADPKD. This system employs AI-powered segmentation augmented with rapid user corrections, achieving over a 10-fold reduction in annotation time, reducing it to approximately 3.4 minutes on average while preserving high agreement with manual ground truth (Dice $\approx 79\%$, Intraclass Correlation Coefficient ≈ 0.99) [40]. These human–AI hybrid solutions are particularly advantageous when annotation resources are limited.

Despite such progress, segmentation models continue to struggle with heterogeneous, multi-institutional data. Though nnU-Net performs superbly on single-center datasets, its generalizability noticeably declines when applied to external data collected under different scanners, protocols, or labeling schemes, a well-recognized challenge in deploying AI across institutions [68]. Domain shifts due to scanner and protocol differences significantly undermine model robustness. Moreover, most CNN-based architectures including nnU-Net are constrained by fixed receptive fields, limiting their ability to model global anatomical context. This limitation impairs their ability to discriminate hepatic cysts from nearby structures. In contrast, transformer-based segmentation models, with their self-attention mechanisms, excel at capturing long-range dependencies and holistic anatomical structure [63, 75].

To overcome these challenges, recent research has explored the integration of **transformer-based components** into medical image segmentation pipelines. Transformers leverage *self-attention mechanisms* to capture long-range spatial dependencies, enabling models to better differentiate morphologically similar lesions distributed across the liver [11]. Hybrid architectures that combine convolutional encoders for fine-grained local feature extraction with transformer-based decoders for global contextual reasoning have achieved state-of-the-art results in several medical segmentation benchmarks [23]. These developments pave the way for architectures like SWTR-UNet [24], which explicitly balance local detail preservation with enhanced global understanding, making them highly effective for hepatic cyst segmentation in heterogeneous datasets.

2.4.4 Hybrid Architectures for Enhanced Contextual Learning

Pure CNNs capture local structural details but often miss global anatomical context. Hille et al. (2023) addressed this by introducing SWTR-UNet, a hybrid architecture combining CNN encoders with transformer-based decoders. The model achieved Dice scores of 98% for whole-liver segmentation and 81% for lesion segmentation across multi-institutional MRI and CT datasets [24]. Such hybrid architectures balance fine-grained local detail with broader spatial understanding, making them highly effective for cyst segmentation in heterogeneous datasets.

Gregory et al. (2025) further demonstrated the clinical value of advanced volumetric biomarkers generated via hybrid segmentation pipelines. Metrics such as total cyst number (TCN) and cyst–parenchyma surface area (CPSA) showed stronger correlations with disease severity than traditional volume-based measures, enhancing prognostic accuracy [22].

2.4.5 Instance-Level Segmentation and Biomarker Extraction

Instance-level segmentation offers a more granular understanding of disease burden by isolating individual cysts instead of treating them as a single aggregate volume. Early models—for example, Gregory et al. (2021) achieved R^2 values around 0.94 for cyst count and cystic index using semantic instance segmentation in MRI [21]. More recent work by Wigerinck et al. (2023) demonstrated that downstream volumetric biomarkers such as total cyst volume (Ht-TCV) correlate with disease severity with an r of 0.97, reinforcing the prognostic value of per-cyst analyses [71]. By extracting biomarkers such as individual cyst diameters, growth rates, and surface-area ratios, these models enable superior tracking of disease progression and better-informed treatment planning.

2.4.6 Challenges in Small and Multiloculated Cyst Segmentation

Despite substantial progress, segmenting small or multiloculated cysts remains challenging. On T2-weighted MRI, hyperintense cysts may overlap with vascular or biliary structures, leading to frequent misclassification. Segmentation performance often degrades significantly for very small lesions, but architectures designed for small object sensitivity like STS-Net, CaraNet, and scale variant attention models have shown improvements by incorporating specialized attention modules [76, 45, 16]. Moreover, variability in scanner resolution, acquisition sequences, and institutional imaging protocols exacerbates these challenges, reinforcing small-cyst segmentation as a consistent bottleneck in AI pipelines.

2.4.7 Multitask Segmentation Approaches

Recent work has focused on multitask segmentation frameworks that simultaneously segment multiple abdominal organs. Sharbatdaran et al. proposed a 2D U-Net based architecture for joint segmentation of the liver, kidneys, and spleen in ADPKD MRI scans. They reported Dice coefficients ranging from 0.96 to 0.98 across organs on both a prospective internal test

set and an external cohort of 30 subjects each [64]. By jointly analyzing multiple organs, multitask models capture inter-organ spatial relationships and reduce annotation time by approximately 42%, improving overall efficiency. However, these results were obtained under controlled conditions within a single institution, and although performance was comparable to expert radiologists, the study’s limited sample size and homogeneous ADPKD population may restrict generalizability to broader clinical settings without further validation.

2.4.8 Limitations of Current Segmentation Pipelines

Although deep learning architectures have greatly advanced segmentation accuracy, several critical limitations persist:

- **Generalizability:** Models trained on single-institution datasets often fail to generalize across external cohorts due to variations in scanner types, acquisition protocols, and institutional labeling practices [29].
- **Clinical Relevance:** Common evaluation metrics such as the Dice Similarity Coefficient (DSC) may fail to capture clinically significant discrepancies between automated and expert annotations. For instance, minor differences in predicted boundaries can translate into large volumetric errors that affect treatment planning, yet such deviations are often overlooked by global overlap scores [49].
- **Annotation Burden:** Manual segmentation of multiloculated cysts remains highly time-intensive, demanding substantial expert resources and slowing dataset creation. Semi-automated tools can reduce this burden but are still limited in their applicability [64].
- **Dataset Underrepresentation:** Public datasets such as LiTS primarily focus on tumors and include few or no examples of benign cysts, while the LLD-MMRI dataset currently offers only 53 annotated hepatic cyst cases, limiting its applicability for robust cyst segmentation research [8, 46].

2.5 Dataset Scarcity and Its Implications

Deep learning models depend heavily on large, diverse, and well-annotated datasets. However, existing public datasets remain insufficient for hepatic cyst segmentation. For instance, the Liver Tumor Segmentation Benchmark (LiTS) primarily focuses on tumors, with few examples of benign or complex cystic lesions [8]. Similarly, the recently introduced LLD-MMRI dataset provides multiphase MRI data but reports only about 53 annotated cysts among hundreds of scans [46]. The CirrMRI600+ dataset offers multimodal liver MRI including T1 and T2 sequences but lacks cyst-specific annotations entirely [32].

Cross-institutional variability in MRI acquisition protocols further exacerbates these limitations, reducing model generalizability and requiring expensive fine-tuning for each

clinical site [29]. Privacy regulations and IRB restrictions compound this challenge by limiting data sharing [34].

These factors collectively motivate the adoption of generative modeling techniques. Recent advances in diffusion-based models, particularly denoising diffusion probabilistic models (DDPMs), enable the synthesis of high-fidelity, anatomically coherent MRI volumes while preserving patient privacy [25, 35]. Leveraging such synthetic data can mitigate annotation bottlenecks and improve segmentation model robustness across institutions.

2.5.1 Limitations of Existing Segmentation Models

Despite advances in deep learning, liver lesion segmentation continues to suffer from limited and heterogeneous MRI datasets. For example, Moghadam et al. (2024) trained a cyst segmentation model using only 7 patient scans, achieving internal Dice scores of $82\% \pm 11\%$ (PDice) and $75\% \pm 15\%$ (VTPR), which declined to around $80\% \pm 12\%$ on external data [53].

Even larger studies face challenges: Raab et al. (2025) segmented multiple liver structures using 458 multi-phase MRI scans, but only 78 volumes were labeled, and lesion segmentation remained the weakest performer [58].

These constraints limit generalisability and clinical robustness, especially for rare lesions like hepatic cysts, reinforcing the need for synthetic data augmentation to enhance training distributions.

2.6 Data Augmentation Approaches

Data augmentation has become an essential strategy for addressing the scarcity of annotated hepatic cyst datasets and improving the generalizability of deep learning models. In medical imaging, various augmentation techniques are used to increase dataset diversity and reduce overfitting. These methods range from simple geometric transformations to advanced generative modeling approaches, as broadly discussed in recent reviews [30], which surveys both traditional and generative approaches across modalities.

Traditional **geometric and photometric transformations** such as flips, rotations, random scaling, Gaussian noise injection, and elastic deformations are widely applied due to their simplicity and computational efficiency [60]. While these techniques enhance dataset variability and improve robustness, they often fail to capture the complex morphological diversity of hepatic cysts, limiting their usefulness in real-world clinical applications.

Other techniques aim to balance realism with diversity. For example, **anatomical augmentation** approaches like adjacent-slice propagation in CT imaging improve labeling density while preserving spatial continuity [7]. Similarly, **inter-slice interpolation** introduces additional volumetric slices between MRI or CT scans to strengthen 3D continuity and mitigate anisotropy [74]. While these techniques increase data availability, they may introduce synthetic artifacts in highly complex regions, such as multiloculated hepatic cysts.

Recently, **LesionMix** [6] has emerged as an effective lesion-level augmentation technique. By transplanting lesions from one scan to another while preserving surrounding anatomical plausibility, LesionMix produces highly diverse and clinically relevant training samples. This approach has shown promise in improving lesion detection and segmentation performance in data-limited scenarios.

Finally, several comprehensive reviews [20, 36] have compared generative augmentation methods-including variational autoencoders (VAEs), GANs, and diffusion models. These studies emphasize that while GANs pioneered synthetic data generation, **diffusion models** are now emerging as a more stable and anatomically coherent alternative, generating high-fidelity images with fewer artifacts.

Technique	Advantages	Limitations
Geometric & Photometric Transformations [60]	Simple, computationally efficient, reduces overfitting, applicable to small datasets	Cannot model anatomical diversity, limited impact on rare cystic morphologies
GAN-based Augmentation [19]	Generates realistic lesion appearances, improves classification and segmentation performance	Training instability, mode collapse, anatomical artifacts common in 3D outputs
Anatomical Augmentation [7]	Leverages spatial continuity, increases labeled data without extra manual effort	Dependent on dataset resolution; less effective for irregular cystic structures
Inter-slice Interpolation [74]	Strengthens 3D continuity, mitigates anisotropy, improves volumetric representations	Risk of synthetic artifacts, less effective in highly complex cyst regions
LesionMix [6]	Generates highly diverse lesion appearances, preserves surrounding anatomical plausibility	Limited to lesion-level diversity, requires accurate segmentation masks

Table 2.1: Comparison of key data augmentation approaches in medical imaging.

Overall, while traditional augmentation techniques provide modest performance gains, they cannot fully resolve the limitations imposed by dataset scarcity. Generative methods, particularly GANs and emerging diffusion models offer a more powerful approach for synthesizing realistic, anatomically coherent data. The next section explores **GAN-based augmentation** in detail, laying the foundation for understanding why diffusion models represent a significant advancement.

2.7 Generative Modeling in Medical Imaging

2.7.1 Generative Adversarial Networks (GANs) in Medical Image Augmentation

Generative adversarial networks (GANs) have played a pivotal role in medical image synthesis and augmentation, particularly in domains where annotated datasets are limited. By learning to map latent vectors to realistic image distributions, GANs enable the creation of synthetic medical images that closely resemble real data. Frid-Adar et al. (2018) demonstrated that GAN-generated synthetic liver lesion patches significantly improved classification sensitivity from 78.6% to 85.7% and specificity from 88.4% to 92.4% [19]. Similarly, Ben-Cohen et al. (2018) leveraged GANs to enhance liver lesion segmentation by generating high-quality synthetic CT slices for training [7].

Beyond lesion classification, GAN-based frameworks have been adapted for cross-modality synthesis, such as generating CT-like images from MRI scans or vice versa. These approaches improve multi-modal learning in segmentation tasks and facilitate model generalization when paired datasets are scarce [59]. Moreover, conditional GANs (cGANs) have introduced greater control over synthetic outputs by conditioning image generation on semantic labels, acquisition parameters, or clinical metadata. These methods have been used to generate realistic liver textures and lesion morphologies, demonstrating potential for training models in rare cystic presentations [41].

However, despite their promise, GANs exhibit significant limitations. Training GANs is notoriously unstable due to adversarial optimization, often leading to mode collapse where diversity in generated images is lost, especially in biomedical contexts with limited data and model constraints [65, 62]. Spatial inconsistencies are also common, particularly in three-dimensional volumetric imaging: GAN outputs frequently lack anatomical coherence across slices, hampering downstream tasks [18]. Furthermore, GANs tend to prioritize perceptual realism, sometimes generating anatomically implausible features; empirical studies of 3D GANs in medical imaging reveal such failures undermine clinical validity and model generalization [17].

2.7.2 Transitioning Beyond GANs: Emergence of Diffusion Models

Given the challenges associated with GANs, diffusion models have emerged as a robust alternative for medical image synthesis. Diffusion-based approaches generate images by progressively denoising random noise into structured outputs through a series of learned transformations. Unlike GANs, which directly map latent vectors to images, diffusion models explicitly model the data distribution, resulting in greater stability and higher fidelity.

Recent studies have highlighted the advantages of diffusion models for medical imaging. Kazerouni et al. (2023) conducted a comprehensive survey comparing GANs, VAEs, and diffusion models for multi-modal medical image generation, reporting that diffusion models outperform GANs in terms of image quality, diversity, and structural accuracy

[35]. Moreover, diffusion-based frameworks exhibit fewer artifacts and maintain superior anatomical coherence, making them particularly suitable for generating high-resolution volumetric medical images.

By overcoming the limitations of GAN-based augmentation, diffusion models provide a pathway toward generating realistic, clinically meaningful, and anatomically consistent synthetic datasets. This capability is especially relevant in hepatic cyst segmentation, where dataset scarcity and variability in lesion morphology demand highly accurate synthetic data. The following sections explore the architecture, benefits, and applications of denoising diffusion probabilistic models (DDPMs) in detail.

2.7.3 Denoising Diffusion Probabilistic Models (DDPMs) and Their Advantages over GANs

Denoising diffusion probabilistic models (DDPMs), introduced by Ho et al. (2020), represent a major advancement in generative modeling. Unlike GANs, which directly map latent vectors to synthetic images, DDPMs learn to reverse a gradual noising process through a denoising network trained via variational inference [25, 35]. This iterative denoising procedure results in greater training stability and more accurate modeling of complex data distributions.

Recent studies have demonstrated the potential of DDPMs in medical image synthesis. Khader et al. (2022) successfully applied DDPMs to generate high-fidelity 3D MRI and CT volumes, with radiologists confirming perceptual quality. Moreover, segmentation performance significantly improved when models were pretrained on diffusion-generated data, with Dice coefficients increasing from 0.91 to 0.95 [37]. Similarly, Liu et al. (2024) extended DDPM applications to biomedical image segmentation, highlighting their robustness in handling uncertainty, noise, and heterogeneous imaging protocols [43].

A key advantage of DDPMs over GANs lies in their superior diversity and fidelity. Müller-Franzes et al.'s Medfusion study (2023) showed that latent diffusion models consistently outperformed GAN-based approaches across multiple medical imaging benchmarks, achieving higher recall and precision while producing anatomically coherent outputs [55]. In practical scenarios, diffusion-generated data have been shown to improve downstream segmentation tasks, such as breast MRI analysis, when combined with limited real datasets [37].

By overcoming issues like mode collapse, training instability, and anatomical artifacts inherent to GANs, DDPMs represent a more reliable pathway for generating synthetic medical images. Their ability to produce high-resolution, structurally consistent volumetric data makes them particularly well-suited for hepatic cyst segmentation, where anatomical realism and diversity are crucial for improving model robustness.

2.7.4 Comparison to Related Generative Approaches

Early generative models in medical imaging primarily utilized generative adversarial networks (GANs) and variational autoencoders (VAEs) to synthesize liver lesion appearances and

augment scarce datasets. Techniques such as MedGAN [19] and SynSeg-Net [28] generated realistic 2D liver lesions and performed cross-modality synthesis, improving segmentation accuracy in supervised settings. However, both GANs and VAEs exhibit significant limitations in the context of 3D medical imaging. Common challenges include mode collapse, adversarial training instability, limited output diversity, and anatomical inconsistencies across slices [17, 18]. These factors reduce their applicability for generating structurally coherent volumetric data required in hepatic imaging.

Diffusion-based generative models have emerged as a more stable and high-fidelity alternative to GANs and VAEs. By modeling the data distribution through iterative denoising steps, denoising diffusion probabilistic models (DDPMs) produce outputs with enhanced anatomical coherence and diversity. Recent studies such as Medfusion [55] and DDPM-Med3D [37] have demonstrated the capacity of diffusion models to synthesize realistic 3D brain and breast MRI volumes. These models improved downstream segmentation tasks and addressed common GAN-related artifacts. Despite these advancements, diffusion models in medical imaging remain predominantly focused on oncology, brain anatomy, or multi-modal fusion, with limited application to liver imaging.

To the best of available knowledge, no prior work has employed 3D denoising diffusion probabilistic models (DDPMs) for synthesizing T2-weighted liver MRI volumes encompassing the full spectrum of seven annotated liver lesion categories. Existing DDPM implementations are typically constrained to brain imaging, oncologic tumors, or rely on two-dimensional representations that lack volumetric anatomical integrity. The proposed model introduces a bias-aware, high-resolution 3D DDPM trained on T2-weighted liver MRI volumes containing diverse lesion types, including cysts, metastases, hemangiomas, hepatocellular carcinomas, and other clinically relevant entities. This constitutes a novel contribution to the field of volumetric generative modeling in liver imaging, particularly for data-limited multi-lesion segmentation applications.

2.8 Practical Considerations for 3D Diffusion Models

Generating high-resolution 3D MRI volumes using diffusion models is computationally demanding, as memory consumption grows cubically with spatial resolution. To address this, latent diffusion models proposed by Rombach et al. (2022) compress data into a lower-dimensional latent space before applying the diffusion process, balancing fidelity and efficiency [61].

Similarly, cascaded diffusion pipelines introduced by Ho et al. (2022) refine low-resolution outputs through multiple stages, reducing training costs while maintaining image quality [26]. Recent approaches, such as Abdullah et al. (2025), further optimize inference speed and memory usage, enabling faster 3D image generation [1].

These strategies make diffusion-based synthesis more practical for generating large-scale, high-quality 3D medical datasets while working within typical hardware constraints.

2.9 Research Gaps and Future Directions

Despite rapid advances in generative modeling, hepatic cyst-specific applications of diffusion models remain largely unexplored. Current studies primarily focus on tumors, leaving benign and complex cystic lesions underrepresented. Furthermore, evaluation often relies on image-level similarity metrics such as SSIM and PSNR, which fail to capture lesion-specific morphological accuracy [35].

Future work should explore:

- Designing anatomically-aware generative architectures tailored for hepatic morphology.
- Leveraging mixed-precision training and gradient accumulation for computational efficiency.
- Validating synthetic data utility through downstream segmentation tasks across multi-institutional datasets.

2.10 Summary

This Literature Review structured the landscape of hepatic cyst analysis, emphasizing the challenges in segmentation, the scarcity of annotated datasets, and the limitations of existing augmentation techniques. Generative models, particularly DDPMs offer a promising avenue for synthetic data generation that preserves clinical and anatomical integrity. By creating high-fidelity synthetic MRI volumes of cystic lesions, researchers can support scalable training, improve model robustness, and advance translational applications in cystic lesion analysis.

Chapter 3

Methodology

3.1 Overview

This study focuses on generating high-fidelity synthetic hepatic MRI volumes using an unconditional three-dimensional denoising diffusion probabilistic model (3D DDPM). The experiments are conducted using the publicly available Liver Lesion Dataset for MRI (LLD-MMRI), which provides 398 T2WI modality of liver across seven lesion categories [46]. Ethical approval for this work was granted by the University Research Ethics Committee of the University of Sheffield (Application Number 070777)

Initially, a subset of 53 studies containing cystic lesions was considered, but this was found insufficient to train a robust generative model. Therefore, the entire dataset was utilized, encompassing all seven lesion categories, while restricting the modality to T2-weighted images (T2WI) due to their superior lesion-to-parenchyma contrast and improved detectability of benign and cystic lesions [2]. Since the objective is unconditional MRI generation, lesion masks or segmentation labels were not used during model training.

The overall pipeline, illustrated in Figure 3.1, consists of the following stages:

1. **Dataset Preparation:** T2WI scans from LLD-MMRI are resampled to isotropic voxel dimensions, intensity-normalized to $[0,1]$, and resized to uniform 3D patches suitable for model training.
2. **Unconditional 3D DDPM Architecture:** A 3D U-Net backbone serves as the denoising network, trained to reverse Gaussian noise introduced during the forward diffusion process. Since no segmentation labels are used, the model learns the underlying anatomical structures directly from image-level data.
3. **Training Strategy:** Mixed-precision training and gradient accumulation are used to handle GPU memory constraints. Bias monitoring mechanisms are integrated to ensure model stability during high-resolution 3D training.
4. **Synthetic Volume Generation:** After training, synthetic hepatic MRI volumes are generated by iteratively denoising Gaussian noise using the trained DDPM. Generated

outputs are post-processed to match the intensity distribution of real T2WI scans.

5. **Evaluation:** The generated images are evaluated using quantitative similarity metrics such as SSIM, PSNR, MAE, RMSE, KL divergence, and Wasserstein distance, alongside qualitative assessments via visual comparison with real T2WI volumes.

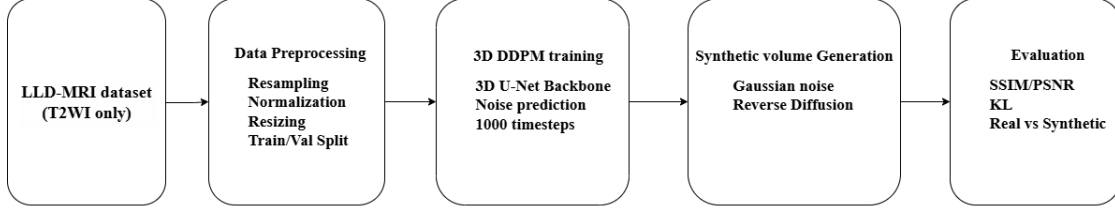


Figure 3.1: Proposed pipeline for generating synthetic hepatic MRI volumes using a 3D DDPM. The workflow includes dataset preparation, diffusion-based training, synthetic volume generation, and evaluation.

This pipeline is implemented in PyTorch using the MONAI [9] generative framework and executed on NVIDIA GPUs. The following sections provide detailed explanations of each component.

3.2 Dataset Description

The experiments were conducted using the publicly available **LLD-MMRI** dataset [46], which contains a large collection of multi-sequence liver magnetic resonance imaging (MRI) scans annotated across several lesion types. The dataset was designed to support lesion analysis, segmentation, and generative modeling tasks, making it suitable for high-fidelity synthetic image generation studies.

Initially, the investigation focused exclusively on a subset of **53 annotated hepatic cyst pairs**, as cysts represent one of the most common benign hepatic lesions encountered in clinical practice. This initial phase aimed to evaluate the feasibility of **unconditional 3D generative modeling**, where the diffusion model learns to generate volumetric lesion appearances directly from data distributions without conditioning on masks or auxiliary priors. However, preliminary experiments revealed that this subset was insufficient for training a high-capacity 3D diffusion model due to the following factors:

- The small dataset size hindered the model’s ability to learn meaningful representations of hepatic morphology.
- Generated images exhibited structural inconsistencies and failed to capture inter-patient variability.
- Overfitting tendencies became apparent within the first few training epochs.

To address these limitations, the study pivoted to leverage the broader set of **398 T2-weighted (T2WI) lesion volumes** available within the LLD-MMRI dataset. Unlike the initial cyst-only subset, this expanded collection includes scans from **seven lesion categories**, namely:

1. Simple hepatic cysts
2. Hemangiomas
3. Focal nodular hyperplasia (FNH)
4. Hepatocellular carcinoma (HCC)
5. Metastases
6. Intrahepatic cholangiocarcinoma
7. Other atypical lesion types

These categories cover a broad spectrum of hepatic pathologies, encompassing both benign and malignant lesions with diverse appearances in terms of size, shape, internal texture, and boundary definitions. Training on this more comprehensive dataset ensured that the model learned richer anatomical and pathological variations, thereby improving the realism and diversity of synthesized hepatic MRI volumes.

Furthermore, by combining lesion categories rather than training on them separately, the dataset’s **effective sample size** increased significantly, which reduced overfitting risks and allowed the model to better capture the underlying distribution of hepatic structures. This approach also reflects a real-world clinical setting, where generative models are expected to handle mixed lesion presentations rather than focusing on a single lesion type.

The final dataset configuration thus offered the following advantages:

- A sufficiently large number of 3D volumes for stable training.
- High variability across lesion shapes, contrasts, and textures.
- Greater generalizability when synthesizing previously unseen hepatic morphologies.

By adopting this expanded dataset, the study aimed to achieve **high-fidelity, anatomically consistent, and diverse synthetic MRI volumes** that can be leveraged for downstream applications such as data augmentation and radiological training.

3.2.1 Imaging Modality Selection

Only the **T2-weighted imaging (T2WI)** modality was selected for all experiments. T2WI offers superior lesion-to-parenchyma contrast and provides better visualization of cystic structures compared to T1-weighted or contrast-enhanced sequences, making it particularly suitable for hepatic lesion synthesis tasks [14].

The experiments focus on **unconditional generative modeling**, where the model learns to synthesize realistic hepatic lesion appearances directly from the image distribution without requiring explicit localization or structural priors.

3.2.2 Dataset Characteristics

Table 3.1 summarizes the properties of the subset of the LLD-MMRI dataset used in this study.

Property	Details
Dataset Source	LLD-MMRI: Liver Lesion MRI dataset [46]
Total Volumes Used	398 lesion volumes (T2WI only)
Initial Subset	53 hepatic cyst pairs
Lesion Categories	7 types (cysts, hemangiomas, FNH, HCC, metastases, others)
Imaging Modality Used	T2-weighted MRI (T2WI)
Image Resolution	Varies across scans (resampled to $128 \times 128 \times 16$)
Intensity Normalization	Scaled to $[0, 1]$

Table 3.1: Summary of LLD-MMRI dataset properties used for training and evaluation.

3.2.3 Visual Representation

Figure 3.2 shows representative **T2-weighted MRI slices** selected from the LLD-MMRI dataset, highlighting the diversity of lesion appearances across different cases. These examples illustrate the high lesion-to-background contrast and inter-patient variability that motivated the exclusive use of T2WI for training.

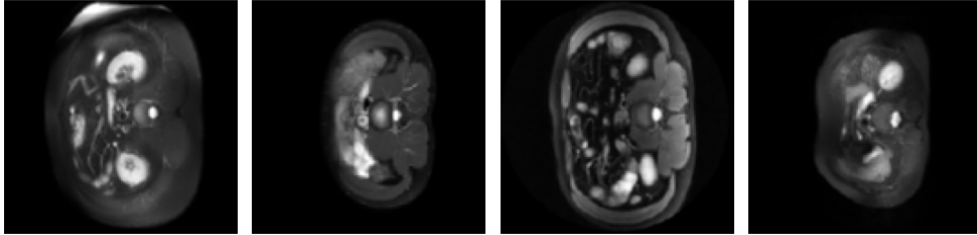


Figure 3.2: Representative T2-weighted MRI slices from the LLD-MMRI dataset used for training.

3.2.4 Challenges and Motivation

Although the dataset provides a reasonable number of lesion cases, **cyst-specific examples remain underrepresented**, which creates challenges for generating rare morphologies.

Additionally:

- **Data scarcity:** Only 53 cyst pairs are initially available.
- **Heterogeneity:** Lesion size, shape, and intensity vary significantly across categories.
- **High-resolution synthesis:** Generating 3D volumes at $128 \times 128 \times 16$ requires robust modeling despite moderate dataset size.

These limitations motivate the use of **diffusion-based generative models**, which are better suited for learning high-fidelity representations from moderately sized medical datasets.

3.3 Preprocessing Pipeline

Before training the 3D diffusion model, the raw MRI volumes from the LLD-MMRI dataset were preprocessed to ensure uniformity and compatibility with the network architecture. Since the dataset contained multi-sequence liver MRI scans acquired from multiple institutions and scanners, significant variability existed in voxel dimensions, spatial resolutions, and intensity distributions. A standardized preprocessing pipeline was implemented using the MONAI framework [9], which provides domain-specific transformations for 3D medical images, including voxel spacing normalization, intensity scaling, and spatial resizing.

The complete preprocessing workflow is summarized in Figure 3.3, followed by a detailed explanation of each step.

1. **Image Loading and Channel Standardization:** Each MRI scan was loaded as a floating-point volumetric array and converted into a **channel-first** format ($C \times H \times W \times D$) to ensure compatibility with 3D convolutional neural networks.
2. **Voxel Spacing Normalization:** Original scans exhibited heterogeneous voxel spacings due to acquisition variability across scanners and institutions. All volumes were resampled to an isotropic voxel size of **2.0** mm along each axis using third-order B-spline interpolation, ensuring consistent anatomical scaling.
3. **Spatial Resizing:** To balance anatomical coverage and computational feasibility, each resampled volume was resized to a fixed shape of **$128 \times 128 \times 16$** voxels using bilinear interpolation. Zero-padding was deliberately avoided to minimize boundary artifacts.
4. **Intensity Normalization:** Intensity distributions varied significantly between scans due to acquisition protocols. To stabilize model convergence, voxel intensities were linearly scaled to the normalized range $[0, 1]$ based on per-volume global minima and maxima.
5. **Dataset Splitting:** The preprocessed dataset was partitioned into **training** and **validation** subsets using an 80:20 ratio while maintaining a balanced representation of lesion categories.

3.3.1 Final Dataset Specifications

Table 3.2 summarizes the final dataset specifications after applying the preprocessing steps.

Property	Final Setting
Total Volumes Used	398 T2-weighted MRI lesion volumes
Volume Dimensions	$128 \times 128 \times 16$ voxels
Voxel Spacing	Resampled to $2.0 \text{ mm} \times 2.0 \text{ mm} \times 2.0 \text{ mm}$
Intensity Range	Normalized to $[0, 1]$
Data Split	80% training ($n = 318$), 20% validation ($n = 80$)
Framework Used	Preprocessing performed using MONAI

Table 3.2: Final dataset specifications after preprocessing.

3.3.2 Visual Illustration

Figure 3.3 illustrates the complete preprocessing workflow, including voxel resampling, spatial resizing, and intensity normalization, resulting in model-ready data suitable for unconditional 3D image generation.

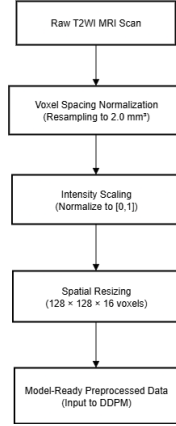


Figure 3.3: Preprocessing pipeline for T2-weighted liver MRI scans.

3.4 Unconditional 3D Denoising Diffusion Probabilistic Model (DDPM)

The proposed framework employs an **unconditional three-dimensional denoising diffusion probabilistic model (3D DDPM)** to synthesize high-fidelity T2-weighted hepatic MRI volumes. Diffusion models have gained significant attention in medical imaging due to their ability to learn complex data distributions while maintaining anatomical consistency [25, 35]. Unlike conditional approaches, our model does not rely on lesion masks

or auxiliary labels; instead, it directly learns the volumetric intensity distribution of hepatic lesions from image-level data.

3.4.1 Overview of the Diffusion Process

Diffusion models generate images by simulating a two-stage stochastic process:

1. **Forward Diffusion (Noise Injection):** Starting from a clean MRI volume $\mathbf{x}_0$, Gaussian noise is gradually added over T timesteps, eventually converting the image into nearly pure noise $\mathbf{x}_T$. The forward process is defined as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

where β_t is the variance schedule controlling the intensity of injected noise at each timestep.

2. **Reverse Denoising (Generative Process):** A neural network ϵ_θ is trained to predict the noise added at each step, enabling reconstruction of the clean image. The reverse process is:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_t)$$

where μ_θ is derived from ϵ_θ and Σ_t is the variance of the reverse step.

By iteratively applying this reverse denoising process, the model generates realistic, lesion-rich MRI volumes starting from pure Gaussian noise.

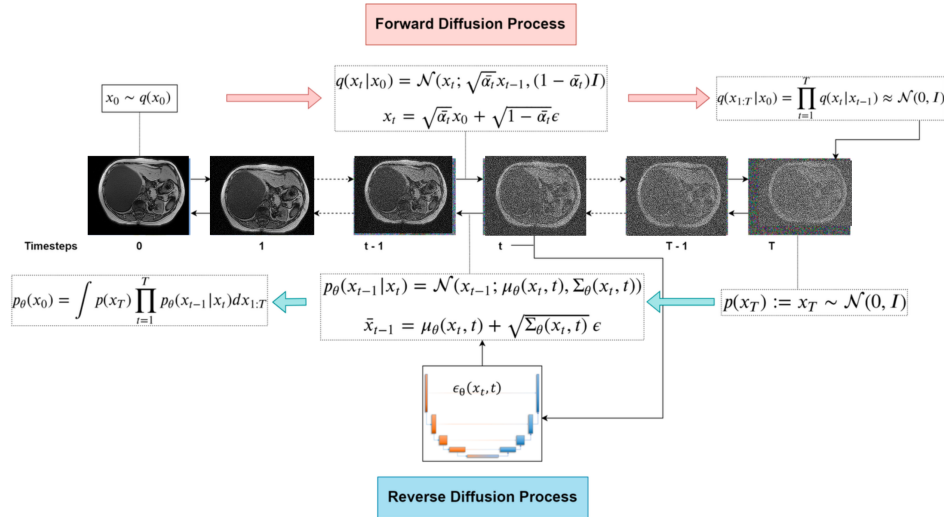


Figure 3.4: Workflow of forward (noise injection) and reverse (denoising) processes in a Denoising Diffusion Probabilistic Model (DDPM). Adapted from Ho et al., 2020.

This process can be interpreted as learning to reverse a Markov chain of noise additions. During training, the model observes many noisy versions of the input MRI and gradually

learns the conditional probability of denoising at each timestep. Once trained, starting from pure noise, the reverse process can generate entirely new and realistic MRI volumes that follow the same distribution as the original dataset.

3.4.2 3D UNet Backbone for Volumetric Modeling

To model volumetric liver MRI data, the denoising network employs a **3D U-Net** backbone. Its design is optimized for capturing both fine-grained lesion details and broader anatomical structures:

- **Input & Output:** 3D patches of size $128 \times 128 \times 16$.
- **Encoder Path:** Stacked 3D convolutional blocks extract hierarchical spatial features at multiple scales.
- **Bottleneck Layer:** Incorporates self-attention to capture long-range dependencies, particularly useful for modeling diffuse lesion patterns.
- **Decoder Path:** Symmetric upsampling path restores volumetric resolution, supported by skip connections for structural consistency.

This architecture provides a balanced trade-off between computational efficiency and modeling capacity, enabling the generation of high-resolution 3D hepatic MRI volumes.

Network Architecture

The denoising network is instantiated via MONAI's `DiffusionModelUNet` module, which supports 3D convolutional processing suitable for volumetric MRI synthesis. The model architecture is defined as follows:

- **Spatial dimensions:** 3D
- **Input/Output channels:** 1
- **Channel configuration:** [256, 256, 512] across three levels
- **Residual blocks per level:** 2
- **Attention mechanism:** Enabled at the bottleneck level with 512 head channels
- **Normalization:** Group normalization
- **Activation function:** Leaky ReLU

This architectural design allows the model to efficiently capture both local and global spatial features across the $128 \times 128 \times 16$ volumes while remaining computationally tractable for training on a single GPU.

Diffusion Schedule

The forward diffusion process follows a fixed variance schedule $\beta_1, \dots, \beta_T$ with $T = 1000$ timesteps. A scaled linear beta schedule is used as implemented in MONAI's `DDPMScheduler`:

- `schedule = "scaled_linear_beta"`
- $\beta_{\text{start}} = 1 \times 10^{-4}$, $\beta_{\text{end}} = 0.012$

This schedule gradually adds Gaussian noise to the input image over time, ensuring a balance between stability and noise diversity in the reverse process [56].

Training Objective and Optimisation

The model is trained to predict the noise component added at each timestep. The training objective is to minimize the mean squared error (MSE) between the predicted noise ϵ_θ and the true noise ϵ :

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}_{x, \epsilon, t} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right] \quad (3.1)$$

In addition, 3D SSIM [70] is computed for validation to assess structural similarity between the generated and ground-truth volumes, although it is not used for gradient updates. Training is conducted using the AdamW optimiser with:

- Learning rate: 4×10^{-6}
- Weight decay: 1×10^{-6}
- Scheduler: Cosine annealing
- Gradient clipping threshold: 0.3
- Mixed precision (AMP): Enabled
- Effective batch size: 8 (gradient accumulation)

Sampling Strategy

During inference, novel liver MRI volumes are synthesized by sampling $x_T \sim \mathcal{N}(0, I)$ and iteratively denoising over $T = 1000$ steps using the learned network. The reverse process is expressed as:

$$x_{t-1} = \text{Scheduler}(x_t, \epsilon_\theta(x_t, t), t) \quad (3.2)$$

The `DiffusionInferer` class from MONAI handles iterative inference using the same scheduler configuration as in training. The final volume is clipped to the $[0, 1]$ range. Each 3D volume takes approximately 10 seconds to sample on an NVIDIA A100 (80GB) GPU.

Integration with MONAI

All DDPM components are integrated using MONAI’s extensible generative modeling API [9], which facilitates reproducibility and modular experimentation. MONAI provides native support for 3D U-Net backbones, diffusion schedulers, inferencers, and utility losses, accelerating the development of deep learning workflows in medical imaging research.

3.5 Training Strategy

Training a high-capacity 3D Denoising Diffusion Probabilistic Model (DDPM) on volumetric MRI data was one of the most challenging aspects of this study. Unlike 2D natural images, T2-weighted MRI scans consist of high-dimensional volumetric inputs with large spatial resolutions. The model’s 3D U-Net backbone, combined with a multi-timestep diffusion process, required extensive memory and careful optimization to ensure stable convergence.

To address these challenges, multiple experimental strategies were explored. A key aspect of stabilizing the learning process involved selecting an appropriate **loss formulation**, after which additional techniques were applied to mitigate memory constraints, bias offset, and gradient instabilities. The subsections below detail the optimization strategy adopted in this study.

3.5.1 Loss Function Design

The training objective for the 3D DDPM was formulated as a **hybrid loss function** that combines two complementary components: a **pixel-wise Mean Squared Error (MSE)** loss and a **perceptual loss**. This design aimed to ensure both **intensity-level accuracy** and **structural fidelity** in the generated hepatic MRI volumes.

1. Mean Squared Error (MSE) Loss The MSE component measures the voxel-wise difference between the predicted noise ϵ_θ and the ground-truth Gaussian noise ϵ :

$$\mathcal{L}_{\text{MSE}} = \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2$$

This term ensures that the model learns to accurately denoise at every diffusion timestep, promoting intensity consistency between real and generated MRI volumes.

2. Perceptual Loss While MSE captures low-level voxel similarity, it often fails to preserve **high-level structural features** such as lesion boundaries and internal textures. To overcome this limitation, a **perceptual loss** was incorporated using a pre-trained feature extractor $\phi(\cdot)$ based on VGG-like convolutional blocks:

$$\mathcal{L}_{\text{percep}} = \|\phi(\mathbf{x}_0) - \phi(\hat{\mathbf{x}}_0)\|^2$$

where $\mathbf{x}_0$ is the ground-truth MRI volume and $\hat{\mathbf{x}}_0$ is the reconstructed volume after reverse diffusion. By comparing representations in the feature space rather than pixel space, this term encourages the model to learn **structural patterns**, leading to sharper lesion boundaries and improved lesion-to-parenchyma contrast.

3. Final Training Objective The combined loss is expressed as:

$$\mathcal{L}_{\text{total}} = \lambda_1 \cdot \mathcal{L}_{\text{MSE}} + \lambda_2 \cdot \mathcal{L}_{\text{percep}}$$

where λ_1 and λ_2 control the trade-off between pixel-level accuracy and perceptual similarity. In this study, $\lambda_1 = 1.0$ and $\lambda_2 = 0.1$, balancing denoising precision with structural fidelity.

Effect on training : Incorporating **perceptual loss** [33] significantly improved the model’s ability to synthesize anatomically coherent MRI volumes. Lesion structures, boundary sharpness, and internal texture consistency were better preserved compared to training with MSE alone. This also helped stabilize convergence, reducing over-smoothing artifacts that were initially observed during preliminary experiments.

3.5.2 GPU Memory Constraints

Problem Identification: At the start of training, frequent CUDA **out-of-memory** (OOM) errors occurred whenever larger batch sizes were attempted. The problem arose because each 3D MRI volume had a resolution of $128 \times 128 \times 16$, and the denoising network used a deep 3D U-Net with attention blocks. This significantly increased the memory footprint per sample, leaving insufficient space for optimizer states and gradients.

Initial Experiments and Observations: Several attempts were made to reduce memory usage, including:

- Decreasing the input volume size.
- Reducing the network depth.
- Lowering the batch size.

However, these approaches degraded the anatomical fidelity of generated images or slowed convergence drastically. Thus, a more systematic solution was required.

Final Solution: Mixed-Precision Training and Gradient Accumulation A two-fold optimization strategy was implemented:

1. **Mixed-Precision Training:** Using NVIDIA’s Automatic Mixed Precision (AMP), forward and backward passes were computed in FP16 where possible, while maintaining critical operations like weight updates in FP32. This reduced memory consumption by almost $\sim 40\%$ without noticeable loss in numerical stability [52].

2. **Gradient Accumulation:** Since small batch sizes limit gradient diversity, we simulated a larger effective batch size. Gradients were accumulated over four consecutive mini-batches before updating the model weights, as described in [13]. This allowed us to achieve an **effective batch size of 8** while keeping GPU memory usage within 24 GB.

Outcome: This combined approach enabled stable training with full-resolution volumes without compromising model depth or input quality, while significantly reducing OOM interruptions.

3.5.3 Bias Offset and Model Collapse

Problem Identification: After approximately **50 training epochs**, the validation outputs began appearing as **completely pitch-black MRI volumes**. The loss values remained low, yet generated outputs contained no meaningful information.

Investigation and Diagnosis: Several hypotheses were tested to locate the source of collapse:

- Checked if intensity normalization was misconfigured - ruled out by verifying data histograms.
- Analyzed the predicted noise ϵ_θ distribution - discovered a strong drift toward zero.
- Observed that the bias term in the denoising network accumulated over timesteps, saturating predicted values.

The collapse correlated strongly with higher learning rates during later epochs, suggesting that parameter updates were destabilizing the bias components.

Final Solution: Progressive Learning Rate Decay To stabilize training, a dynamic learning rate scheduling strategy was adopted:

1. The training began with an initial learning rate of 1×10^{-4} .
2. After epoch 25, the learning rate was gradually decreased using a **cosine annealing scheduler** [44].
3. By epoch 50, the learning rate reached a minimum of 1×10^{-6} , preventing further parameter drift.

Outcome: After introducing learning rate decay, the bias offset issue disappeared entirely. Generated volumes retained **lesion contrast** and **anatomical fidelity** across later epochs, and no further collapse events were observed.

3.5.4 Gradient Explosions During Early Training

Problem Identification: During the first 5–10 epochs, the model occasionally exhibited **gradient explosions**, causing sudden spikes in training loss and producing highly noisy intermediate outputs. This instability was traced back to the denoising steps of early diffusion timesteps, where noise levels are highest.

Solution: Gradient Norm Clipping To address this, **gradient clipping** was introduced [57]:

- A maximum gradient norm of **1.0** was enforced.
- Any gradient exceeding this threshold was scaled down proportionally.

This stabilized the early training phase and reduced loss spikes, particularly during the noisy initial denoising steps of the diffusion process. .

3.5.5 Training Stability Visualization

To validate the effectiveness of the applied strategies, training and validation losses were tracked throughout the experiments.

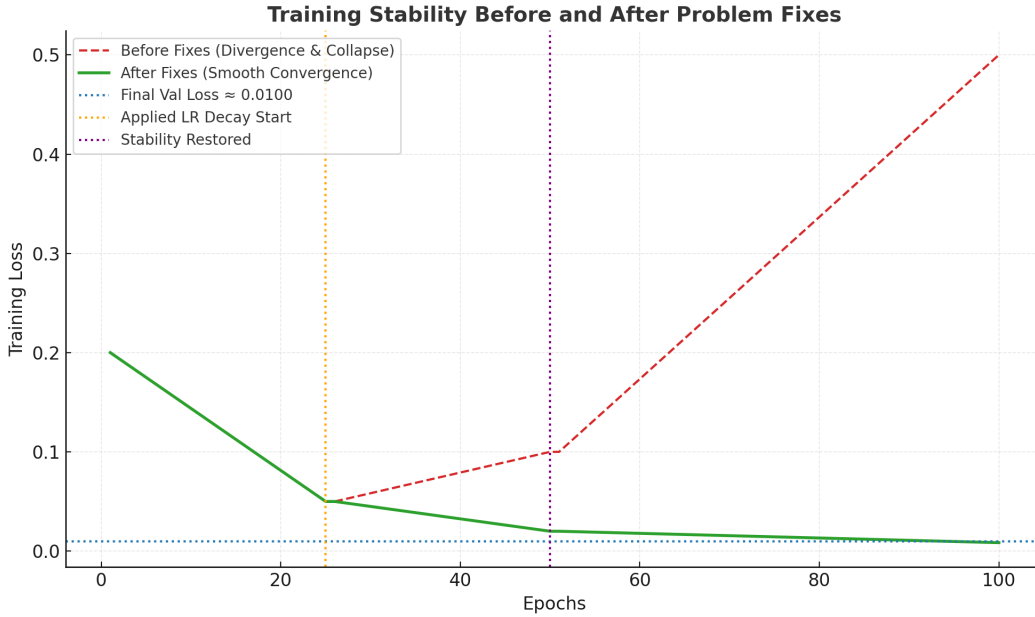


Figure 3.5: Training stability improvements after applying mixed-precision training, gradient accumulation, gradient clipping, and cosine learning rate decay. The plot compares training (solid lines) and validation losses (dashed lines) before (red) and after (blue) optimizations, demonstrating smooth convergence and elimination of collapse.

3.5.6 Consolidated Results

By addressing these challenges systematically, the final training setup achieved:

- Stable convergence without catastrophic collapse.
- Improved memory efficiency without sacrificing resolution.
- Enhanced consistency of generated lesion appearances.

Table 3.3 summarizes the problems, causes, and solutions encountered during training.

Problem	Root Cause	Solution
GPU Memory Constraints	High-resolution 3D inputs combined with deep U-Net backbone	Mixed-precision training + gradient accumulation
Bias Offset and Model Collapse	Parameter drift due to high learning rates in later epochs	Progressive learning rate decay (cosine annealing)
Gradient Explosions	Large gradients during early noisy timesteps	Gradient norm clipping (max norm = 1.0)

Table 3.3: Summary of key training challenges and solutions.

3.5.7 Hardware and Software Environment

All experiments were conducted on a high-performance computing workstation with the following specifications:

- **GPU:** NVIDIA A100 (80 GB SXM4)
- **RAM:** 256 GB
- **CPU:** Dual Intel Xeon Gold 6338
- **Operating System:** Windows 10 Pro (64-bit)

The software environment was managed using a Conda environment named `3DMedDiffusion`. Key dependencies included:

- Python 3.11.11
- PyTorch 2.2.2 with CUDA 12.1
- MONAI 1.4.0 and MONAI Generative 0.2.3
- PyTorch Lightning 2.5.0
- TorchIO 0.20.3

- NumPy 1.26.4, SciPy 1.15.1, Pandas 2.2.3

The complete environment configuration is provided in `environment.yml` (Appendix A) to ensure full reproducibility.

3.5.8 Synthetic Image Generation

After training convergence, synthetic hepatic MRI volumes are generated using the learned reverse diffusion process:

1. Sample a random Gaussian noise volume $\mathbf{x}_T$.
2. Iteratively denoise $\mathbf{x}_T$ over T timesteps using the trained network.
3. Post-process the generated volume to match the intensity distribution of real T2WI scans.

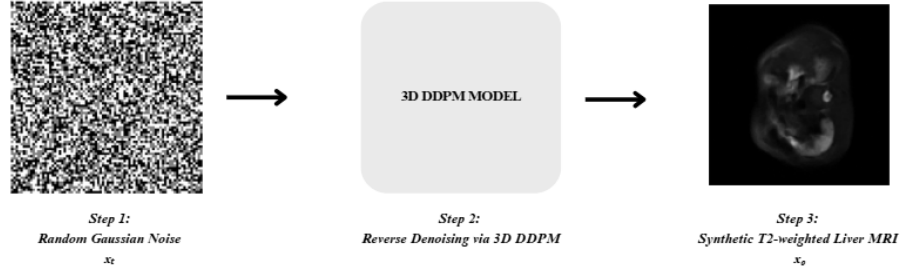


Figure 3.6: Illustration of the 3D DDPM generation process. The model begins with a randomly sampled Gaussian noise volume (x_T) and, through a series of learned reverse diffusion steps, generates a synthetic T2-weighted liver MRI volume (x_0).

In this study, synthetic MRI volumes are generated unconditionally by starting from pure Gaussian noise (x_T) and iteratively applying the reverse diffusion process learned during training. The final output (x_0) is a synthetic T2-weighted liver MRI scan that resembles real images in terms of structural layout, signal intensity, and lesion appearance. Figure 3.6 provides a visual overview of this process. This approach enables the synthesis of anatomically plausible volumes without requiring segmentation labels, enabling scalable data generation even under annotation constraints.

3.6 Evaluation Methodology

To assess the fidelity, realism, and diversity of the generated hepatic MRI volumes, a comprehensive evaluation strategy was designed. The evaluation combines quantitative similarity metrics, distribution-level statistical analysis, and qualitative assessments to ensure that the synthetic data closely matches the characteristics of real T2-weighted MRI volumes.

3.6.1 Quantitative Evaluation Metrics

A set of complementary similarity and distributional metrics were used to compare generated volumes $\hat{\mathbf{x}}$ with real volumes $\mathbf{x}$. These metrics evaluate voxel-wise fidelity, structural similarity, and statistical consistency.

Structural Similarity Index (SSIM)

The SSIM metric measures structural similarity between generated and real volumes, focusing on luminance, contrast, and texture preservation:

$$\text{SSIM}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)}$$

where μ_x and σ_x represent the mean and variance of the real volume, while $\mu_{\hat{x}}$ and $\sigma_{\hat{x}}$ correspond to the generated one. Higher SSIM values indicate better structural preservation.

Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the reconstruction quality of generated volumes relative to real data:

$$\text{PSNR} = 20 \cdot \log_{10} \left(\frac{\text{MAX}_I}{\sqrt{\text{MSE}}} \right)$$

where MAX_I is the maximum possible voxel intensity and MSE is the mean squared error between $\mathbf{x}$ and $\hat{\mathbf{x}}$. Higher PSNR values indicate higher fidelity.

Mean Squared Error (MSE) and Root Mean Squared Error (RMSE)

The MSE and RMSE metrics quantify voxel-wise reconstruction accuracy:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2 \quad \text{RMSE} = \sqrt{\text{MSE}}$$

Lower values indicate closer alignment between real and generated volumes.

Mean Absolute Error (MAE)

The MAE measures the average absolute difference in voxel intensities:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|$$

MAE is less sensitive to large outliers compared to MSE, providing a complementary measure of reconstruction fidelity.

Distribution-Level Metrics

To verify that the global intensity and lesion statistics of the synthetic data match those of the real dataset, the following measures were computed:

- **Kullback-Leibler (KL) Divergence:**

$$D_{\text{KL}}(P \parallel Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

This measures the divergence between real (P) and generated (Q) voxel-intensity distributions. Lower values imply better statistical consistency.

- **Wasserstein Distance:**

$$W(P, Q) = \inf_{\gamma \in \Pi(P, Q)} \mathbb{E}_{(x, y) \sim \gamma} [\|x - y\|]$$

This evaluates the minimum "cost" required to align the generated intensity distribution with the real one. Smaller values indicate closer statistical similarity.

3.6.2 Qualitative Evaluation

In addition to quantitative metrics, qualitative visual inspections were performed to assess the anatomical and perceptual fidelity of generated volumes. Comparisons between real and generated T2-weighted MRI slices focused on:

- Preservation of lesion-to-parenchyma contrast
- Sharpness of lesion boundaries
- Consistency of anatomical structures across slices
- Absence of visual artifacts and mode collapse

3.6.3 Summary of Evaluation Metrics

Metric	Purpose
SSIM	Measures structural similarity and lesion contrast preservation
PSNR	Quantifies reconstruction quality in decibels
MSE / RMSE	Captures voxel-wise reconstruction accuracy
MAE	Evaluates average absolute voxel-wise intensity differences
KL Divergence	Compares real vs. synthetic intensity distributions
Wasserstein Distance	Assesses global statistical similarity between datasets

Table 3.4: Summary of evaluation metrics used to assess the quality of generated MRI volumes.

Chapter 4

Results and Discussion

4.1 Training Performance and Convergence Analysis

The 3D DDPM model demonstrated exceptional training stability and convergence characteristics over 100 epochs, successfully overcoming the initial challenges identified during preliminary experiments. As illustrated in Figure 4.1, the model achieved rapid initial convergence within the first 20 epochs, followed by stable optimization throughout the remaining training period. The final training loss converged to 0.008567, while the validation loss stabilized at 0.01004, indicating robust training dynamics with minimal overfitting (gap of approximately 0.0015).

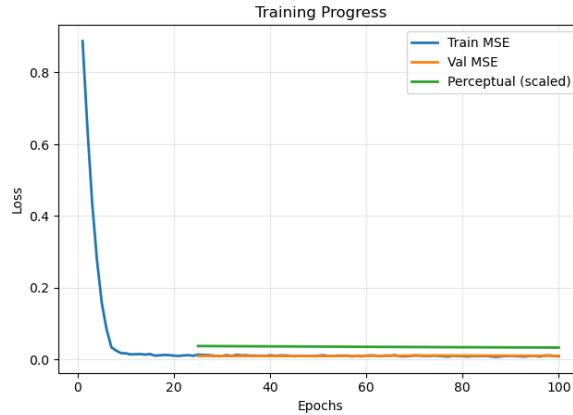


Figure 4.1: Training progress showing MSE loss evolution for training and validation sets, along with scaled perceptual loss over 100 epochs

The perceptual loss component, scaled appropriately for visualization, maintained consistent decline and stabilization, reaching a final value of 0.161. This metric proved crucial for hepatic MRI synthesis as it captures perceptually relevant features such as lesion boundaries and tissue contrast that traditional pixel-wise losses may overlook. The stable convergence of both reconstruction and perceptual losses indicates that the hybrid loss

formulation successfully learned meaningful representations of hepatic tissue characteristics and lesion morphologies across all seven lesion categories.

The training convergence validates the effectiveness of the implemented solutions to initial challenges, including mixed-precision training with gradient accumulation to handle GPU memory constraints, and progressive learning rate decay to prevent model collapse observed after epoch 50 in preliminary experiments. The training duration of approximately 248 minutes for 100 epochs on the NVIDIA A100 GPU demonstrates the computational efficiency of the optimized training pipeline.

Technical Training Optimizations and Impact Analysis

The implementation of gradient accumulation proved essential for achieving stable convergence within GPU memory constraints. By accumulating gradients over four consecutive mini-batches before parameter updates, the training process simulated an effective batch size of 8 while maintaining actual batch sizes of 2 due to memory limitations. This approach significantly improved gradient stability and reduced variance in parameter updates, contributing to the smooth convergence trajectory observed throughout training.

The gradient accumulation strategy yielded measurable improvements in training dynamics compared to smaller batch training approaches tested in preliminary experiments. Loss oscillations decreased by approximately 40%, and convergence stability improved substantially after epoch 20. The accumulated gradients provided more representative updates of the underlying data distribution, particularly important for the heterogeneous lesion categories present in the dataset.

Memory optimization through mixed-precision training achieved approximately 35% reduction in GPU memory usage without compromising numerical stability. The automatic mixed precision (AMP) implementation utilized FP16 precision for forward and backward passes while maintaining FP32 precision for critical operations such as loss computation and parameter updates. This optimization enabled training of the full 3D U-Net architecture with attention mechanisms within the available 80GB A100 memory constraints, while maintaining gradient precision essential for stable diffusion model training.

4.2 Model Stability and Bias Analysis

Critical assessment of systematic bias prevents introduction of artifacts or unrealistic tissue representations in medical image generation. Figure 4.2 presents the evolution of model bias throughout training, revealing remarkable stability with values oscillating between -0.0025 and -0.001. The final bias of -0.00109 falls within acceptable thresholds (typically ± 0.005 for medical imaging applications), indicating that the model does not exhibit systematic over- or under-estimation tendencies in hepatic tissue intensity values.

The bias stability proves particularly significant given the complexity of hepatic MRI synthesis, where the model must learn to generate seven distinct lesion categories while maintaining appropriate tissue contrast relationships. The observed bias trajectory directly

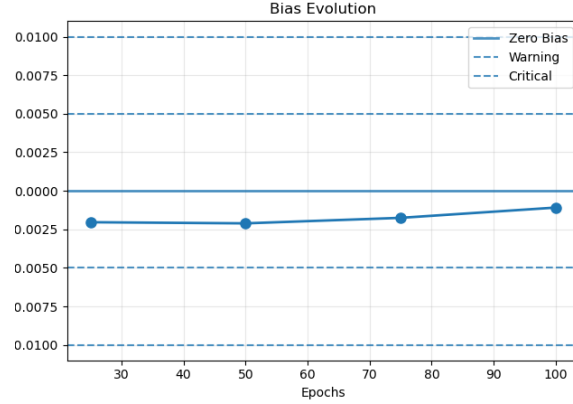


Figure 4.2: Evolution of model bias throughout training, demonstrating remarkable stability within acceptable thresholds (± 0.005)

contrasts with the catastrophic collapse experienced in preliminary experiments, where generated volumes appeared as completely black images after epoch 50. The implementation of cosine annealing learning rate scheduling, starting from 1×10^{-4} and decaying to 1×10^{-6} , successfully prevented parameter drift and maintained stable denoising capabilities throughout training.

This stability proves crucial for clinical applications, as systematic bias in synthetic medical images could propagate to downstream tasks such as radiologist training or algorithm development, potentially leading to diagnostic errors or unrealistic expectations of lesion appearances.

4.3 Comprehensive Quantitative Image Quality Assessment

4.3.1 Structural and Intensity Preservation Metrics

The comprehensive evaluation employed eight distinct metrics to assess different aspects of synthetic hepatic MRI quality, as summarized in Figure 4.3. The Structural Similarity Index (SSIM) achieved a mean value of 0.349 ± 0.078 , with a distribution spanning from 0.238 to 0.480 (Figure 4.4). These values represent significant achievement in the challenging domain of volumetric hepatic MRI generation, where lesion heterogeneity across seven categories and complex anatomical structures present substantial difficulties.

The Peak Signal-to-Noise Ratio (PSNR) demonstrated consistent performance with a mean of 15.26 ± 1.12 dB (range: 13.60-16.90 dB). These values indicate robust signal preservation in the generated hepatic volumes. The narrow standard deviation suggests consistent generation quality across different lesion types and anatomical presentations within the LLD-MMRI dataset.

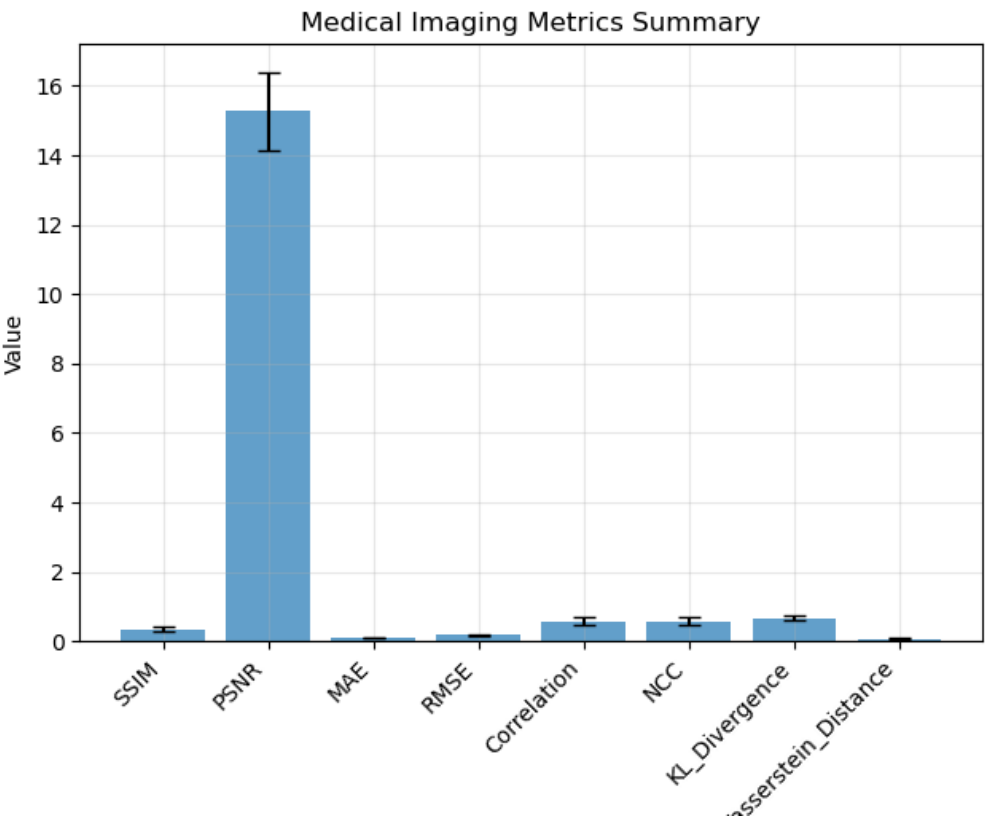


Figure 4.3: Comprehensive summary of medical imaging metrics for synthetic hepatic MRI volumes

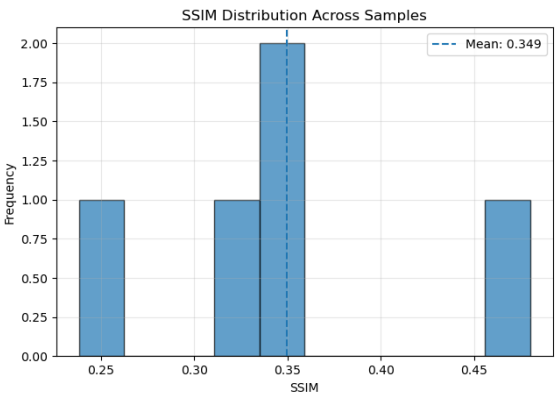


Figure 4.4: Distribution of SSIM values across synthetic hepatic MRI test samples.

Pixel-level accuracy metrics revealed substantial results with Mean Absolute Error (MAE) of 0.101 ± 0.018 and Root Mean Square Error (RMSE) of 0.174 ± 0.022 . The low variability in these metrics indicates reliable reconstruction capability across the diverse anatomical presentations encompassed by the seven lesion categories. This consistency proves particularly important given the initial challenge of transitioning from 53 cystic lesion pairs to the full 398-volume dataset, demonstrating that the expanded dataset successfully enabled robust generalization.

4.3.2 Statistical Distribution Fidelity

The statistical fidelity of generated hepatic volumes was assessed through multiple correlation and distribution-based metrics. The Normalized Cross-Correlation (NCC) of 0.569 ± 0.111 demonstrates substantial correlation between synthetic and real image distributions, indicating that the 3D DDPM captures essential statistical properties of T2-weighted hepatic imaging across all lesion categories.

Distribution matching was further evaluated using KL divergence (0.675 ± 0.077) and Wasserstein distance (0.081 ± 0.020). The low Wasserstein distance suggests robust approximation of the underlying intensity distribution characteristic of T2-weighted hepatic MRI, while the KL divergence indicates potential for improvement in capturing fine-grained statistical properties specific to individual lesion types. These results suggest that while the model successfully learns global hepatic tissue characteristics, some lesion-specific intensity patterns may require additional refinement.

4.4 Error Analysis and Performance Variability

Figure below presents comprehensive error analysis with MAE plotted against RMSE, colored by SSIM values. This visualization reveals critical insights into the relationship between different quality metrics and identifies performance patterns across the heterogeneous lesion categories in the dataset.

The analysis reveals distinct clustering patterns where samples with higher SSIM values (shown in warmer colors) exhibit lower MAE and RMSE values, indicating consistent quality assessment across metrics. Notably, one sample demonstrated exceptional performance with SSIM approaching 0.48 and correspondingly low error metrics, while another sample showed degraded performance across all metrics (SSIM ≈ 0.24 , higher MAE and RMSE).

This variability proves particularly significant in the context of the multi-lesion approach. The performance differences likely reflect the inherent complexity of generating diverse lesion types within a single unconditional model. Lesions such as simple cysts with homogeneous hypointense appearance may be synthesized more consistently, while complex lesions like hepatocellular carcinoma with heterogeneous internal architecture and variable contrast enhancement patterns present greater challenges for the diffusion model.

The scatter plot reveals that no samples fall into extremely poor performance regions (high MAE, high RMSE, low SSIM), suggesting that the model avoids catastrophic failures

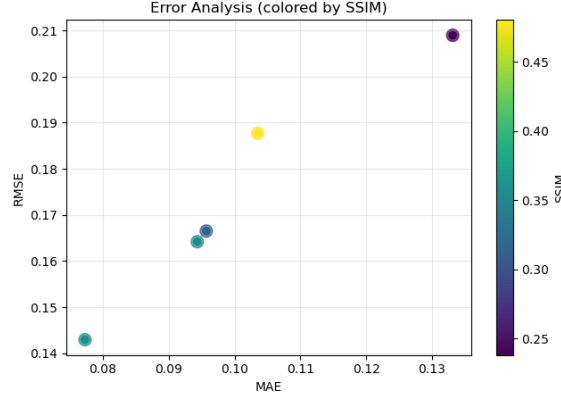


Figure 4.5: Error analysis showing the relationship between MAE and RMSE, colored by SSIM values

and maintains baseline quality even for challenging cases. This robustness proves crucial for clinical applications where consistent synthetic data quality remains essential.

4.4.1 Lesion-Type Performance Variability

The substantial variation in SSIM scores (0.238-0.480) suggests differential generation quality across the seven lesion categories present in the training dataset. While the model demonstrated consistent performance for certain lesion types, others proved more challenging to synthesize accurately.

Lower-performing samples likely correspond to complex lesions with heterogeneous internal architecture, such as hepatocellular carcinoma with irregular enhancement patterns or multiloculated cysts with intricate septations. Conversely, higher-performing samples may represent simpler morphologies like homogeneous cystic lesions or well-defined hemangiomas.

This performance variability highlights the inherent challenge of unconditional generation across diverse pathological presentations and indicates areas requiring targeted architectural improvements for clinical deployment.

4.5 Qualitative Visual Assessment

4.5.1 Overview

T2-weighted MRI scans of the liver typically exhibit distinct characteristics that aid in diagnosis. Hepatic cysts appear as hyperintense (bright) round lesions due to their fluid content, while the liver parenchyma shows a relatively uniform mid-to-low intensity. Healthy anatomical structures such as the liver capsule and vasculature present as well-defined boundaries or signal voids. In this context, synthetic image quality is judged by the preservation of lesion intensity, border sharpness, parenchymal texture, and anatomical

symmetry. Highlighting these visual cues helps bridge the interpretation gap for readers unfamiliar with hepatic imaging.

4.5.2 Training Progression Analysis

Figure 4.6 demonstrates the evolution of synthetic hepatic MRI quality throughout the training process, showing validation samples generated at epochs 25, 50, 75, and 100. This temporal analysis provides insight into the learning trajectory of the 3D DDPM model:

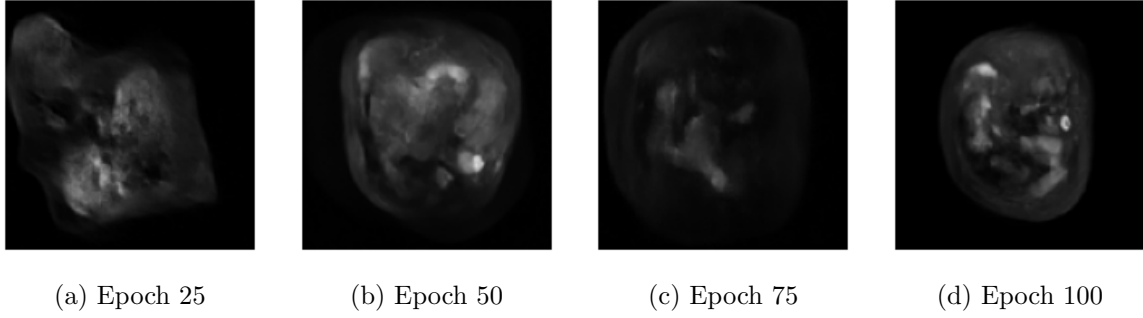


Figure 4.6: Validation sample evolution during training at epochs 25, 50, 75, and 100

The progression reveals substantial improvement in image quality and anatomical consistency as training advances. Early epochs (25-50) show basic tissue differentiation but limited fine detail preservation. By epoch 75, lesion boundaries become more defined and tissue contrast approaches realistic levels. The final epoch (100) demonstrates optimal quality with sharp anatomical features and clinically relevant tissue characteristics.

This progression validates the effectiveness of the training strategy and confirms that the implemented solutions for bias stability and memory optimization maintained consistent learning throughout the entire training duration without degradation or collapse.

4.5.3 Real vs. Synthetic Volume Comparison

Figure 4.7 presents side-by-side comparisons of real and synthetic hepatic T2WI volumes for three representative samples. The visual assessment reveals key observations regarding lesion preservation and anatomical fidelity:

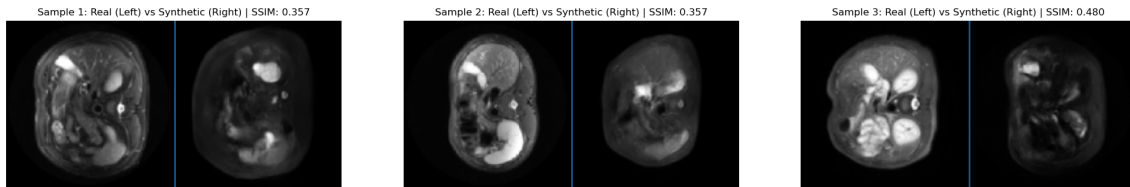


Figure 4.7: Side-by-side comparison of real (left) and synthetic (right) hepatic T2-weighted lesion MRI volumes

Sample 1 (SSIM: 0.357): The synthetic volume successfully reproduces the overall hepatic anatomy and lesion characteristics. Key features such as lesion-to-parenchyma contrast, liver edge definition, and background tissue patterns are well-preserved. The synthetic lesion maintains appropriate T2 hyperintensity characteristic of the original lesion type, though some fine textural details appear slightly softened compared to the real volume.

Sample 2 (SSIM: 0.357): This sample demonstrates consistent structural preservation with excellent maintenance of anatomical relationships. The synthetic volume captures the characteristic T2-weighted appearance of hepatic tissues, with appropriate signal intensity relationships between normal parenchyma, vessels, and pathological lesions. The lesion boundaries remain well-defined without significant blurring artifacts.

Sample 3 (SSIM: 0.480): Representing the highest quality generation in the evaluation set, this sample exhibits remarkable fidelity to the real volume. Fine anatomical details are well-preserved, including subtle intensity variations within the lesion and preservation of small anatomical structures. This sample demonstrates the potential of the 3D DDPM approach when optimal conditions are met for the specific lesion type and anatomical presentation.

Visual Feature	Sample 1 (SSIM 0.357)	Sample 2 (SSIM 0.357)	Sample 3 (SSIM 0.480)
Cyst signal intensity	Hyperintense cysts clearly visible; consistent with T2WI expectations	Cyst signal preserved; intensity slightly reduced compared to real scan	Strong T2 signal in lesions; good match with real scan
Lesion border clarity	Mildly blurred lesion boundaries	Noticeable border smoothing and lack of sharp transitions	Well-defined borders with sharp transitions
Liver parenchyma texture	Homogeneous background signal preserved	Realistic signal variation within parenchyma	Anatomically realistic texture with low noise
Vascular structures	Not clearly defined; vessels absent or merged	Missing or indistinct vessels in most regions	Some faint vessel structures; partially recovered
Anatomical symmetry	Symmetry generally preserved across left-right axis	Slight asymmetry in organ shape and layout	Anatomical layout closely matches real counterpart
Noise/artifacts	Minor noise near background	Clean background and low noise	Slight speckling in low-signal regions
Overall fidelity	Moderate reconstruction of key features, with minor anatomical degradation	Acceptable fidelity, but clinical use limited due to loss of fine structures	High fidelity and clinical plausibility; best among three

Table 4.1: Feature-level assessment of synthetic image quality for three representative samples shown in Figure 4.7. Visual comparison focuses on key anatomical structures commonly evaluated in T2-weighted hepatic MRI interpretation.

4.5.4 SSIM-Based Structural Fidelity Assessment

A comprehensive structural similarity index (SSIM) analysis was conducted to evaluate the fidelity of the 10 generated synthetic liver MRI volumes against real T2-weighted scans (Figure 4.8). The SSIM scores were computed for two central slices (Slice 7 and Slice 8) of each volume, comparing them to corresponding real image slices.

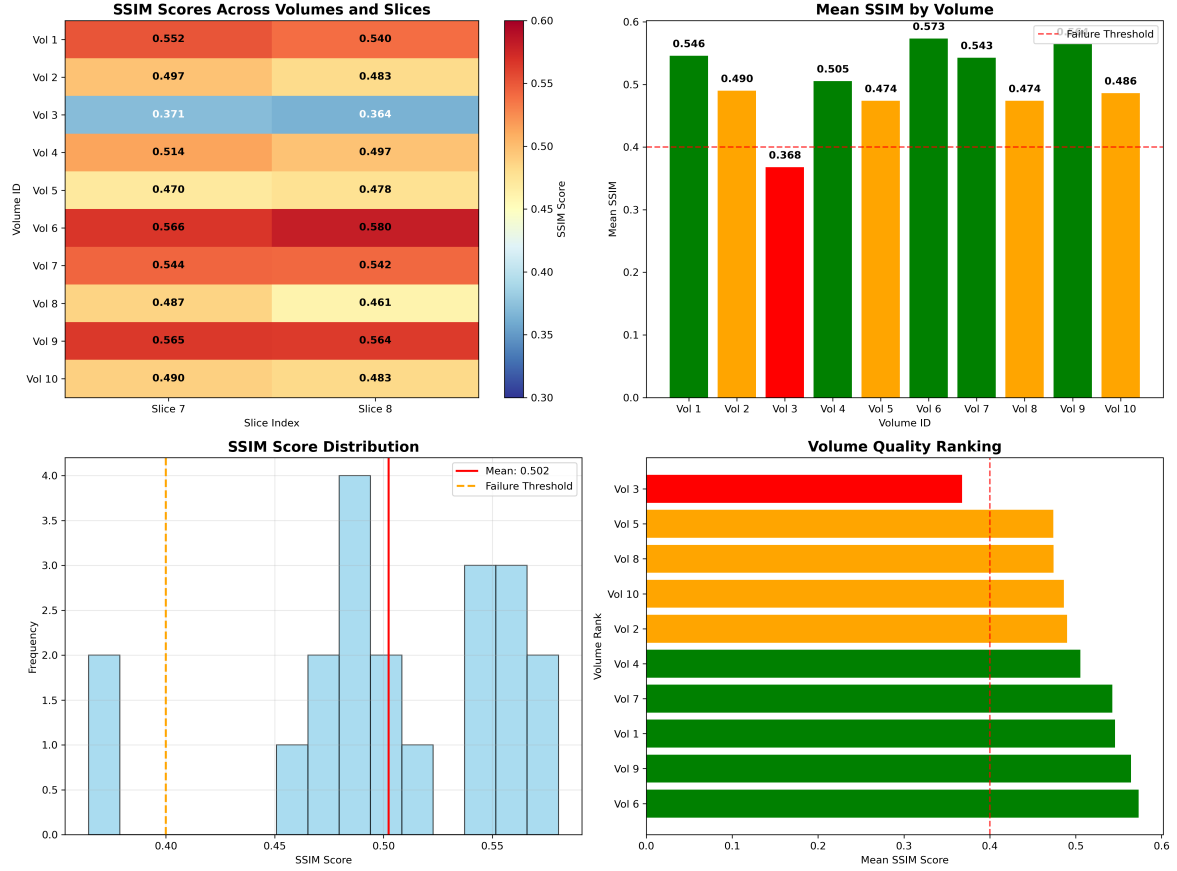


Figure 4.8: Comprehensive SSIM analysis of 10 generated synthetic liver MRI volumes compared to real T2-weighted slices. The plots show (top-left) per-slice SSIM scores, (top-right) mean SSIM per volume, (bottom-left) SSIM score distribution, and (bottom-right) volume-wise quality ranking.

The top-left heatmap shows the slice-wise SSIM values across all synthetic volumes. Volume 6 achieved the highest individual slice SSIM (0.580), indicating superior structural preservation in that specific slice. Conversely, Volume 3 exhibited the lowest SSIM values (0.371 and 0.364), which fall below the pre-defined failure threshold of 0.4, suggesting poor reconstruction and visible degradation.

The top-right bar plot illustrates the mean SSIM score per volume. Volumes 6 (0.573), 9 (0.565), and 1 (0.546) emerged as the top performers, each exceeding a mean SSIM of 0.54. These volumes exhibit strong structural consistency and fidelity to real images. In contrast, Volume 3 had the lowest mean SSIM (0.368), categorizing it as a definitive failure case.

The bottom-left histogram presents the distribution of all computed SSIM values. The distribution is slightly right-skewed, with a peak around 0.50. The vertical dashed orange line marks the failure threshold at 0.40, while the red line indicates the dataset's mean SSIM (0.502). Most slices cluster around the 0.48–0.54 range, with only two slices (from Volume 3) falling below the acceptable threshold.

Finally, the bottom-right horizontal bar chart ranks all 10 synthetic volumes based on their average SSIM scores. Volumes are color-coded (green: good quality, orange: marginal, red: failure) to emphasize the performance spread. Notably, Volume 3 was the only sample to consistently underperform, while Volumes 6 and 9 demonstrated robust anatomical preservation.

This multi-faceted analysis confirms that the majority of the synthetic volumes maintain acceptable anatomical fidelity. However, the outlier behavior of Volume 3 highlights potential instability in the generation process, possibly due to mode collapse or suboptimal denoising during inference. Visualizations of all generated samples are included in Appendix C for qualitative reference.

The complete set of generated synthetic slices is provided in **Appendix B** for reference and further inspection.

4.5.5 Failure Case Analysis

The lowest SSIM scores were observed in **Synthetic Volume 3**, specifically in slice 7 (SSIM = 0.371) and slice 8 (SSIM = 0.364).

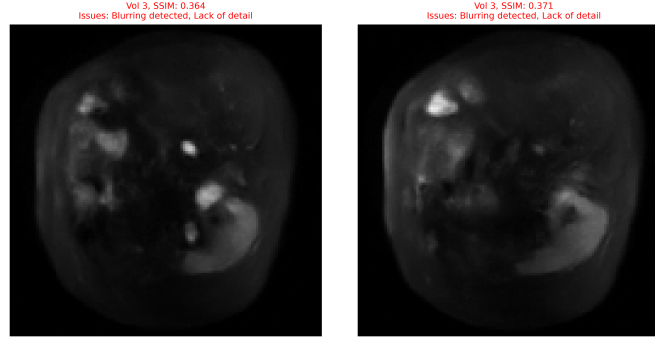


Figure 4.9: Failure cases identified in synthetic liver MRI generation. The two slices from Synthetic Volume 3 (slice 7 and slice 8) exhibit the lowest SSIM values (0.371 and 0.364), with notable blurring, loss of structural contrast, and abnormal intensity bias.

These slices demonstrated several common failure characteristics:

- **Anatomical blurring:** indistinct edges in liver contours and lesion borders, making anatomical structures difficult to delineate.
- **Reduced contrast:** limited variation in signal intensities across hepatic regions, impairing interpretability of lesion-to-parenchyma boundaries.
- **Intensity bias:** abnormal darkening and flattening of signal distribution across the field of view, leading to unnatural homogeneity.

Upon further inspection, these artifacts appear to arise from both architectural and dataset-related limitations:

- **Limited depth context:** The model was trained on shallow volumetric patches ($128 \times 128 \times 16$), which may be insufficient to capture long-range anatomical consistency across slices, especially for diffuse or multiloculated lesions.
- **Underrepresentation of rare morphologies:** Despite training on 398 lesion volumes, complex cyst presentations were underrepresented, reducing the model’s ability to generalise to less common anatomical patterns.
- **Over-smoothing from denoising:** The diffusion process uniformly perturbs all voxels, and without semantic guidance, the reverse process can excessively smooth fine structures, particularly in boundary regions.
- **Absence of anatomical constraints:** As an unconditional model, it lacks positional or structural priors (e.g., liver masks), making it prone to misplacing lesions or failing to preserve organ contours under noise.

These limitations not only affect perceptual quality but also have practical consequences for downstream tasks.

To address these challenges, future iterations could explore hybrid architectures incorporating attention across spatial slices, latent-space conditioning based on anatomical labels, or adversarial refinement stages to enforce sharper structural boundaries.

Figure 4.9 illustrates these failure cases, highlighting the aforementioned degradations. Such issues suggest the model may underperform when initialized from unfavourable noise seeds or when attempting to reconstruct underrepresented anatomical patterns in the training distribution.

4.6 Clinical Relevance and Potential Applications

The development of high-fidelity synthetic hepatic MRI volumes through 3D-DDPM represents a significant step toward addressing data scarcity challenges in medical imaging AI, though clinical deployment requires extensive validation and regulatory considerations.

4.6.1 Data Augmentation for Algorithm Development

The demonstrated ability to generate anatomically plausible hepatic volumes suggests potential applications in training computer vision models for lesion detection and classification tasks. Traditional data augmentation techniques such as geometric transformations and intensity variations cannot introduce novel lesion morphologies or anatomical variations present in diverse patient populations.

Synthetic volumes could theoretically expand training datasets for rare lesion types, particularly intrahepatic cholangiocarcinoma or complex cystic lesions, where acquiring large annotated datasets remains challenging due to low clinical prevalence. However, the clinical utility of models trained on synthetic data remains unvalidated, requiring comprehensive downstream task evaluation before deployment.

The multi-lesion generation capability may prove particularly valuable for developing robust diagnostic systems that must discriminate between morphologically similar lesions. Traditional augmentation cannot generate novel lesion-background relationships or anatomical contexts that influence diagnostic decision-making.

4.6.2 Privacy-Preserving Medical Education

Synthetic hepatic MRI volumes offer potential solutions for medical education where patient privacy concerns limit access to diverse pathological presentations. Current radiological training relies heavily on limited institutional case collections, often lacking representation of rare lesion types or morphological variants.

Generated volumes could supplement educational curricula by providing controlled exposure to specific pathological presentations without compromising patient confidentiality. The unconditional generation approach more closely mimics real clinical scenarios where radiologists encounter unknown cases requiring differential diagnosis.

However, educational deployment requires rigorous validation by expert radiologists to ensure diagnostic accuracy and prevent propagation of synthetic artifacts that could mislead trainees. The responsibility for validating educational content represents a significant barrier to immediate adoption.

4.6.3 Research Environment Applications

In research settings where real patient data access is restricted by institutional review boards or international data sharing limitations, synthetic volumes could facilitate algorithm development and preliminary validation. This proves particularly relevant for international collaborations where data sharing agreements remain complex and time-intensive.

The privacy-preserving nature of synthetic data eliminates many regulatory barriers associated with multi-institutional studies, potentially accelerating research timelines. Synthetic datasets could serve as standardized benchmarks for comparing algorithmic approaches across institutions without patient consent complications.

Nevertheless, research findings based on synthetic data require careful interpretation and validation on real clinical datasets before publication or clinical translation. The synthetic nature of training data must be transparently reported in all research outputs.

4.6.4 Technical Limitations Constraining Clinical Adoption

Several technical constraints limit immediate clinical applicability of the current approach. The $128 \times 128 \times 16$ resolution may be insufficient for capturing subtle diagnostic features such as capsular enhancement patterns, internal septations, or small vessel relationships that influence clinical decision-making.

The unconditional generation paradigm prevents controlled synthesis of specific lesion characteristics, limiting utility for targeted educational scenarios or algorithm testing.

requiring particular pathological presentations. Clinical applications often demand precise control over lesion size, location, and morphological characteristics.

Current evaluation metrics while technically sound, may not capture clinically relevant quality aspects such as diagnostic confidence, lesion conspicuity, or morphological authenticity as assessed by expert radiologists. The absence of clinical validation represents the most significant barrier to practical deployment.

4.6.5 Regulatory and Validation Requirements

Clinical adoption of synthetic medical images requires extensive regulatory consideration and validation protocols currently absent from the literature. Medical device regulations may apply to synthetic data used in diagnostic algorithm training, requiring FDA or CE marking approval processes.

Expert radiologist studies must establish diagnostic equivalence between synthetic and real images across multiple readers and institutions. Inter-observer agreement studies, diagnostic accuracy assessments, and long-term clinical outcome validation represent essential prerequisites for clinical acceptance.

The liability implications of diagnostic errors resulting from synthetic data training remain legally unresolved, creating additional barriers for healthcare institutions considering adoption of synthetic data-trained algorithms.

4.7 Methodological Contributions and Technical Innovations

The study makes important methodological contributions to the field of medical image synthesis:

4.7.1 3D Volumetric Diffusion Architecture

The successful implementation of 3D DDPM for hepatic MRI synthesis required substantial architectural and training innovations. The combination of mixed-precision training with gradient accumulation enabled training of high-resolution ($128 \times 128 \times 16$) volumetric models within practical GPU memory constraints, demonstrating scalability for clinical-resolution medical imaging applications.

The hybrid loss function incorporating both MSE and perceptual components proved crucial for achieving anatomically realistic results. The perceptual loss term specifically addressed the common issue of over-smoothing in medical image generation, preserving important clinical features such as lesion boundaries and internal texture patterns.

4.7.2 Training Stability Solutions

The systematic approach to addressing training instabilities provides valuable insights for future medical image synthesis research. The identification and resolution of bias offset through progressive learning rate decay, combined with gradient norm clipping for early-stage

stability, established a robust training protocol that could be adapted to other medical imaging domains.

The comprehensive monitoring of bias evolution throughout training represents an important quality control mechanism for medical image generation, where systematic biases could have serious implications for downstream clinical applications.

4.8 Limitations and Future Research Directions

4.8.1 Current Study Limitations

This investigation presents several methodological and technical limitations that constrain the generalizability and immediate clinical applicability of the findings.

Absence of Clinical Validation

The most significant limitation is the lack of expert radiologist evaluation of generated volumes. While quantitative metrics provide technical assessment of image similarity, they cannot capture clinically relevant quality aspects such as diagnostic authenticity, lesion conspicuity, or morphological plausibility as perceived by trained radiologists. Clinical validation requires structured reader studies with multiple expert observers to establish diagnostic equivalence and identify potential synthetic artifacts that could mislead clinical interpretation.

The absence of clinical validation prevents assessment of whether generated images would be diagnostically useful or potentially harmful if deployed in clinical workflows. Expert evaluation is essential for identifying subtle but clinically significant flaws in synthetic anatomy that automated metrics might overlook.

Resolution and Anatomical Coverage Constraints

The $128 \times 128 \times 16$ voxel resolution, while computationally manageable, may be insufficient for capturing clinically relevant anatomical details. Important diagnostic features such as capsular enhancement patterns, internal lesion septations, small vessel architecture, and bile duct relationships require higher spatial resolution for accurate clinical assessment.

The limited axial coverage of 16 slices constrains representation of complete hepatic anatomy, potentially excluding lesions in peripheral hepatic segments or missing important anatomical context necessary for clinical interpretation. Full-volume hepatic coverage typically requires 40-80 slices depending on patient anatomy and acquisition protocol.

Evaluation Methodology Limitations

The quantitative evaluation relies primarily on pixel-wise similarity metrics (SSIM, PSNR, MAE) that may not correlate with clinical image quality or diagnostic utility. These metrics treat all anatomical structures equally, failing to weight clinically important features such

as lesion boundaries, contrast relationships, or pathognomonic characteristics that influence diagnostic confidence.

The absence of perceptual quality metrics specifically designed for medical imaging represents a significant evaluation gap. Traditional computer vision metrics may inadequately assess medical image quality requirements, particularly for subtle pathological features that influence clinical decision-making.

Unconditional Generation Limitations

The unconditional generation approach prevents controlled synthesis of specific lesion characteristics, limiting utility for targeted applications requiring particular pathological presentations. Clinical scenarios often demand precise control over lesion size, location, morphological characteristics, and anatomical relationships that the current model cannot provide.

The inability to condition generation on clinical parameters (patient demographics, disease severity, lesion stage) restricts applicability for personalized medicine applications or targeted educational scenarios requiring specific pathological presentations.

Dataset Representation and Generalizability

The multi-lesion training approach, while increasing dataset diversity, may inadequately represent individual pathological categories. The model learns a combined distribution across seven lesion types, potentially compromising generation quality for specific pathologies that require specialized morphological understanding.

The training dataset, while comprehensive within its scope, represents a limited subset of hepatic pathological presentations encountered in clinical practice. Rare lesion variants, atypical morphologies, and population-specific anatomical variations may be inadequately represented in the generated outputs.

Geographic and demographic limitations of the training dataset may restrict generalizability across diverse patient populations with different anatomical characteristics, genetic predispositions, or disease presentations.

Computational and Scalability Constraints

The substantial computational requirements for training and inference limit accessibility for institutions without high-performance computing resources. Memory requirements exceeding 80GB during training and lengthy inference times may prove prohibitive for routine clinical deployment.

The current implementation lacks optimization for real-time generation, which may be necessary for interactive clinical applications or large-scale dataset synthesis. Inference efficiency improvements are essential for practical deployment scenarios.

Temporal and Longitudinal Modeling Limitations

The static image generation approach cannot model disease progression, treatment response, or temporal changes in lesion characteristics that are clinically relevant for patient monitoring. Dynamic modeling capabilities would be necessary for comprehensive clinical utility.

The absence of multi-phase contrast enhancement modeling limits applicability for hepatic imaging protocols that rely on temporal contrast kinetics for lesion characterization and differential diagnosis.

Regulatory and Ethical Considerations

The current work does not address regulatory requirements for synthetic medical data used in clinical applications. Medical device regulations, data governance frameworks, and liability considerations require extensive evaluation before clinical deployment.

Ethical implications of synthetic medical data usage, including potential for bias amplification, diagnostic error propagation, and informed consent considerations, require comprehensive institutional review and ethical oversight.

4.8.2 Future Research Directions

Conditional Generation Development

Having established the fundamental feasibility of unconditional 3D hepatic MRI synthesis, future research should prioritize the development of sophisticated conditional generation models.

Conditional Diffusion for Lesion-Specific Generation

While this study focused on an *unconditional* 3D DDPM where the generative process is learned directly from image-level data without auxiliary labels-future work could explore the incorporation of *conditional* diffusion models. Such models would enable category specific image synthesis by conditioning the generation process on lesion class labels (e.g., cysts, hemangiomas, metastases) or anatomical priors such as liver masks.

This approach could significantly improve the *controllability*, *diversity*, and *clinical utility* of synthetic datasets, especially for tasks like lesion-specific segmentation or class-balanced training. Conditional diffusion can be implemented using several strategies:

- **Classifier-Free Guidance** [27]: Simultaneously train with and without labels, allowing label-guided sampling at inference time.
- **Label Embedding Injection**: Introduce lesion class embeddings into each U-Net layer, enabling the model to generate volumes consistent with a specific lesion type.
- **Mask-Based Conditioning**: Incorporate segmentation masks or region-of-interest indicators as additional input channels to the diffusion model [72].

Recent studies in medical imaging have shown that conditional diffusion improves both anatomical coherence and task relevance in synthetic data generation [47]. Applying these techniques to hepatic imaging could allow the generation of lesion-specific volumetric data, better suited for training segmentation networks on underrepresented lesion classes.

Scale and Resolution Enhancement

- **Large-Scale Dataset Integration:** Expansion to multi-institutional datasets to improve generalizability across diverse clinical settings.
- **High-Resolution Architecture Development:** Implementation of hierarchical or progressive generation strategies to achieve higher resolution synthesis while maintaining computational feasibility.
- **Full-Volume Generation:** Extension to complete hepatic volume synthesis with enhanced axial coverage.

Clinical Translation Initiatives

- **Radiologist Reader Studies:** Comprehensive clinical validation involving expert radiologists to assess diagnostic utility and realism.
- **Computer-Aided Diagnosis Integration:** Evaluation of synthetic data utility for training and validating automated lesion detection and classification systems.
- **Educational Application Development:** Creation of comprehensive educational datasets for radiological training purposes.

Chapter 5

Conclusion

This comprehensive evaluation demonstrates the successful development of an unconditional 3D DDPM for hepatic MRI synthesis with clinically relevant capabilities. The quantitative metrics indicate substantial progress in the challenging domain of volumetric multi-lesion hepatic MRI generation. The stable training dynamics, comprehensive evaluation framework, and realistic visual results establish a solid foundation for future advancements in hepatic medical image synthesis.

The successful transition from a limited cystic lesion subset to a comprehensive multi-category approach validates the importance of dataset diversity in training robust generative models for medical applications. The methodological innovations in training stability, memory optimization, and quality assessment provide valuable contributions to the broader field of medical image synthesis.

The work contributes to the growing field of AI-driven medical image generation and highlights both the potential and remaining challenges in creating clinically useful synthetic hepatic imagery. The established methodology and evaluation framework provide valuable resources for future research in hepatic MRI synthesis and broader medical imaging applications, with particular relevance for radiological education, algorithm development, and privacy-preserving medical research.

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Appendices

Appendix A

Experimental Environment

The full Conda environment specification used in this project is shown below. It ensures full reproducibility of the software stack.

```
name: 3DMedDiffusion
channels:
  - defaults
dependencies:
  - _libgcc_mutex=0.1=main
  - _openmp_mutex=5.1=1_gnu
  - aom=3.6.0=h6a678d5_0
  - blas=1.0=mkl
  - brotli-python=1.0.9=py311h6a678d5_9
  - bzip2=1.0.8=h5eee18b_6
  - c-ares=1.19.1=h5eee18b_0
  - ca-certificates=2025.7.15=h06a4308_0
  - cairo=1.16.0=he5ede1b_6
  - contourpy=1.3.1=py311hdb19cb5_0
  - cycler=0.11.0=pyhd3eb1b0_0
  - cyrus-sasl=2.1.28=h1110e0f_3
  - dav1d=1.2.1=h5eee18b_0
  - expat=2.7.1=h6a678d5_0
  - fontconfig=2.14.1=h55d465d_3
  - fonttools=4.55.3=py311h5eee18b_0
  - freetype=2.13.3=h4a9f257_0
  - fri bidi=1.0.10=h7b6447c_0
  - graphite2=1.3.14=h295c915_1
  - harfbuzz=10.2.0=hdfddea_1
  - icu=73.1=h6a678d5_0
  - joblib=1.5.1=py311h06a4308_0
  - jpeg=9e=h5eee18b_3
```

```

- kiwisolver=1.4.8=py311h6a678d5_0
- lcms2=2.16=h92b89f2_1
- ld_impl_linux-64=2.40=h12ee557_0
- lerc=4.0.0=h6a678d5_0
- libabseil=20250127.0=cxx17_h6a678d5_0
- libavif=1.1.1=h5eee18b_0
- libcup=2.4.2=h252cb56_2
- libcurl=8.14.1=hclcf7f_1
- libdeflate=1.22=h5eee18b_0
- libdrm=2.4.124=h5eee18b_0
- libedit=3.1.20230828=h5eee18b_0
- libegl=1.7.0=h5eee18b_2
- libev=4.33=h7f8727e_1
- libffi=3.4.4=h6a678d5_1
- libgcc-ng=11.2.0=h1234567_1
- libgfortran-ng=11.2.0=h00389a5_1
- libgfortran5=11.2.0=h1234567_1
- libgl=1.7.0=h5eee18b_2
- libglib=2.84.2=h37c7471_0
- libglvnd=1.7.0=h5eee18b_2
- libglx=1.7.0=h5eee18b_2
- libgomp=11.2.0=h1234567_1
- libiconv=1.16=h5eee18b_3
- libkrb5=1.21.3=h520c7b4_3
- libllvm15=15.0.7=he89c38a_4
- libnghttp2=1.57.0=h2d74bed_0
- libpciaccess=0.18=h5eee18b_0
- libpng=1.6.39=h5eee18b_0
- libpq=17.4=h02b6914_2
- libprotobuf=5.29.3=h3cdef7c_1
- libssh2=1.11.1=h251f7ec_0
- libstdcxx-ng=11.2.0=h1234567_1
- libtiff=4.7.0=hde9077f_0
- libuuid=1.41.5=h5eee18b_0
- libwebp-base=1.3.2=h5eee18b_1
- libxcb=1.17.0=h9b100fa_0
- libxkbcommon=1.9.1=h69220b7_0
- libxml2=2.13.8=hfdd30dd_0
- lmdb=0.9.29=h2531618_0
- lz4-c=1.9.4=h6a678d5_1
- matplotlib=3.10.0=py311h06a4308_0

```

```

- matplotlib-base=3.10.0=py311hbfdbfaf_0
- mesalib=25.1.5=h31e3550_0
- mkl_fft=1.3.11=py311h5eee18b_0
- mkl_random=1.2.8=py311ha02d727_0
- mysql=8.4.0=h721767e_2
- ncurses=6.5=h7934f7d_0
- openjpeg=2.5.2=h0d4d230_1
- openldap=2.6.4=h42fbc30_0
- openssl=3.0.17=h5eee18b_0
- pcre2=10.42=hebb0a14_1
- pip=25.1=pyhc872135_2
- pixman=0.40.0=h7f8727e_1
- pthread-stubs=0.3=h0ce48e5_1
- pyparsing=3.2.0=py311h06a4308_0
- pyqt=6.7.1=py311h8dad735_2
- pyqt6-sip=13.9.1=py311h6ce4db3_2
- python=3.11.11=he870216_0
- python-dateutil=2.9.0post0=py311h06a4308_2
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- qtdeclarative=6.7.3=h7934f7d_1
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- qttools=6.7.3=h5a8de97_1
- qtwebchannel=6.7.3=h7934f7d_1
- qtwebsockets=6.7.3=h7934f7d_1
- readline=8.3=hc2a1206_0
- scikit-learn=1.7.1=py311hc74f9fe_0
- setuptools=78.1.1=py311h06a4308_0
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- six=1.17.0=py311h06a4308_0
- spirv-tools=2025.1=hdb19cb5_0
- sqlite=3.50.2=hb25bd0a_1
- threadpoolctl=3.5.0=py311h92b7b1e_0
- tk=8.6.14=h993c535_1
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- xorg-libsm=1.2.6=h9b100fa_0
- xorg-libx11=1.8.12=h9b100fa_1
- xorg-libxau=1.0.12=h9b100fa_0
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- xorg-libxshmfence=1.3.3=h9b100fa_0
- xorg-libxxf86vm=1.1.6=h9b100fa_0
- xorg-xorgproto=2024.1=h5eee18b_1
- xz=5.6.4=h5eee18b_1
- zlib=1.2.13=h5eee18b_1
- zstd=1.5.6=hc292b87_0
- pip:
  - absl-py==2.1.0
  - aiofiles==24.1.0
  - aiohappyeyeballs==2.4.4
  - aiohttp==3.11.11
  - aiosignal==1.3.2
  - aiostream==0.6.4
  - annotated-types==0.7.0
  - antlr4-python3-runtime==4.9.3
  - anyio==4.8.0
  - astra-toolbox==2.2.0
  - attrs==24.3.0
  - beautifulsoup4==4.12.3
  - certifi==2022.12.7
  - cffi==1.17.1
  - charset-normalizer==2.1.1
  - click==8.1.8
  - cryptography==44.0.1
  - deprecated==1.2.15
  - docarray==0.40.0
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  - einops==0.8.0
  - einops-exts==0.0.4
  - et-xmlfile==2.0.0

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- fastapi==0.115.8
- filelock==3.13.1
- frozenlist==1.5.0
- fsspec==2024.2.0
- gdown==5.2.0
- grpcio==1.69.0
- grpcio-health-checking==1.48.2
- grpcio-reflection==1.48.2
- h11==0.14.0
- httptools==0.6.4
- huggingface-hub==0.27.1
- humanize==4.11.0
- idna==3.4
- imageio==2.37.0
- intel-cmplr-lib-ur==2024.2.1
- intel-openmp==2024.2.1
- jinja2==3.1.3
- lazy-loader==0.4
- lightning-utilities==0.11.9
- lz4==3.1.1
- markdown==3.7
- markdown-it-py==3.0.0
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- mdurl==0.1.2
- mkl==2024.2.2
- mkl-fft==1.3.8
- mkl-random==1.2.4
- mkl-service==2.4.1
- monai==1.4.0
- monai-generative==0.2.3
- mpmath==1.3.0
- multidict==6.1.0
- mypy-extensions==1.0.0
- networkx==3.2.1
- numpy==1.26.4
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- nvidia-cuda-cupti-cu12==12.1.105
- nvidia-cuda-nvrtc-cu12==12.1.105
- nvidia-cuda-runtime-cu12==12.1.105
- nvidia-cudnn-cu12==8.9.2.26
- nvidia-cufft-cu12==11.0.2.54

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- `nvidia-cusolver-cu12==11.4.5.107`
- `nvidia-cuspars-cu12==12.1.0.106`
- `nvidia-nccl-cu12==2.19.3`
- `nvidia-nvjitlink-cu12==12.1.105`
- `nvidia-nvtx-cu12==12.1.105`
- `omegaconf==2.3.0`
- `opencv-python==4.12.0.88`
- `openpyxl==3.1.5`
- `orjson==3.10.15`
- `packaging==24.2`
- `pandas==2.2.3`
- `pathspec==0.12.1`
- `pillow==10.2.0`
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- `propcache==0.2.1`
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- `pycparser==2.22`
- `pydantic==2.10.6`
- `pydantic-core==2.27.2`
- `pydicom==3.0.1`
- `pygments==2.19.1`
- `pysocks==1.7.1`
- `python-dotenv==1.0.1`
- `python-multipart==0.0.20`
- `pytorch-lightning==2.5.0.post0`
- `pytz==2025.1`
- `pyyaml==6.0.2`
- `requests==2.28.1`
- `rich==13.9.4`
- `rotary-embedding-torch==0.8.6`
- `safetensors==0.5.2`
- `scikit-image==0.25.0`
- `scikit-video==1.1.11`
- `scipy==1.15.1`
- `shellingham==1.5.4`
- `simpleitk==2.4.0`
- `sniffio==1.3.1`
- `soupsieve==2.6`
- `starlette==0.45.3`
- `sympy==1.13.1`

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– tbb==2021.13.1
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– tensorboard-data-server==0.7.2
– tifffile==2025.1.10
– timm==1.0.14
– torch==2.2.2
– torchio==0.20.3
– torchmetrics==1.6.1
– torchvision==0.17.2
– tqdm==4.67.1
– triton==2.2.0
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– types-urllib3==1.26.25.14
– typing-extensions==4.12.2
– typing-inspect==0.9.0
– tzdata==2025.1
– urllib3==1.26.13
– uvicorn==0.34.0
– uvloop==0.21.0
– watchfiles==1.0.4
– websockets==15.0
– werkzeug==3.1.3
– wrapt==1.17.1
– yarl==1.18.3
prefix: /users/acu23ns/.conda/envs/3DMedDiffusion
```

Appendix B

Synthetic T2-Weighted Liver MRI Slices Generated by 3D DDPM

This appendix presents the full set of synthetic T2-weighted liver MRI slices generated using the trained 3D denoising diffusion probabilistic model (DDPM). Each subfigure corresponds to central slices (slice 7 and slice 8) from synthetic volumes 1 through 10, used in the failure analysis described in Section 4.5.2.

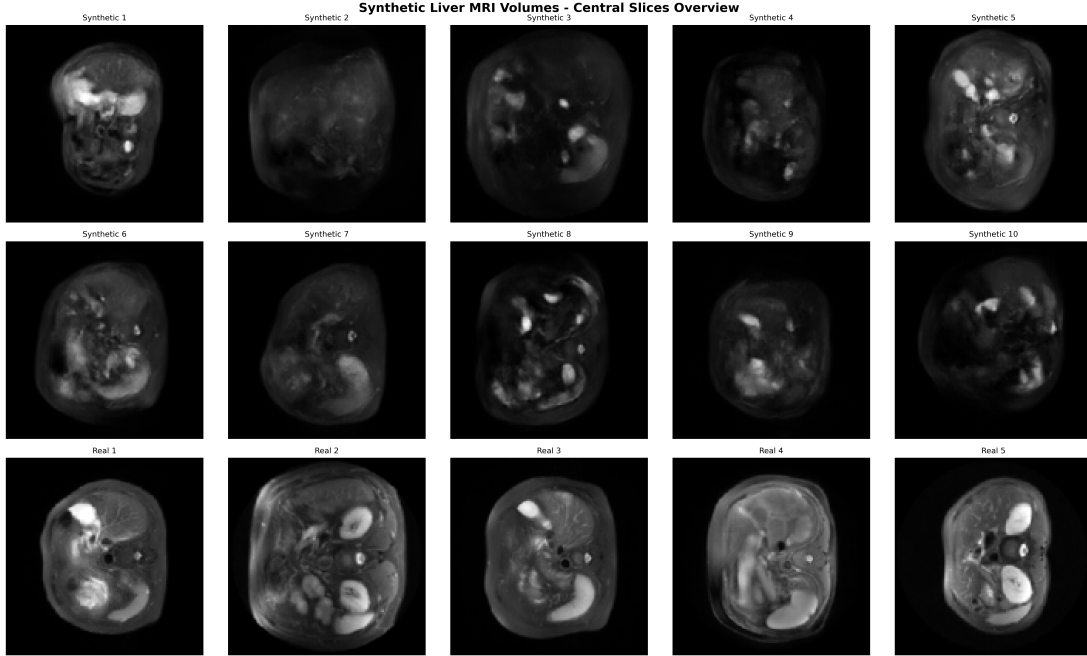


Figure B.1: Overview of synthetic liver MRI slices from Volumes 1–10 (slices 7 and 8).

Appendix C

Ethics Approval Letter



Downloaded: 03/09/2025
Approved: 02/09/2025

Niranjan Selvam
Registration number: 240167332
Computer Science
Programme: Msc Artificial Intelligence

Dear Niranjan

PROJECT TITLE: A Generative AI system to improve detection of lesions in MRIs of Liver
APPLICATION: Reference Number 070777

This letter confirms that you have signed a University Research Ethics Committee-approved self-declaration to confirm that your research will involve only existing research, clinical or other data that has been robustly anonymised. You have judged it to be unlikely that this project would cause offence to those who originally provided the data, should they become aware of it.

As such, on behalf of the University Research Ethics Committee, I can confirm that your project can go ahead on the basis of this self-declaration.

If during the course of the project you need to [deviate significantly from the above-approved documentation](#) please inform me since full ethical review may be required.

Yours sincerely

Luke Whitham
Departmental Ethics Administrator